

Parental Investments and Socio-Economic Gradients in Learning across European Countries*

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Abstract

This paper examines how learning gaps between children of rich and poor parents have evolved over the past two decades in European countries, using data from PISA and PIAAC. We find that Nordic countries – traditionally known for low income inequality – have recently experienced a widening of these learning gaps, which are now as large as those observed elsewhere in Europe. We also show that differences in parental investments between rich and poor families in the Nordic countries are similar to those in other European countries, underscoring the persistent influence of families.

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1 Introduction

Socio-economic status (SES) gradients in learning outcomes, measured by schooling or test scores, emerge early and persist over time (Carneiro and Heckman, 2002; Schady et al., 2015), accounting for a large fraction of the intergenerational transmission of income (e.g., Bolt et al. (2021)).¹ These SES gradients in test scores vary widely across countries, reflecting broader patterns in the intergenerational transmission of inequality. Such inequalities are passed down from parents to children, but the degree of transmission differs significantly across countries (e.g., Corak (2013)) and even across regions within the same country (e.g., Chetty et al. (2014)). Understanding the sources of this variation can inform the design of more effective education and family policies.

In this paper, we ask which countries have high SES gradients in test scores, how these gradients have changed over time, and with what correlates they are associated. We use data from the Program for International Student Assessment (PISA), which collects comparable measures of knowledge in math, reading, and science, as well as socio-economic status (SES), for 15-year-old adolescents in multiple countries (OECD, 2018; OECD, 2023). We document that, surprisingly, SES gradients in learning are not particularly high or low in countries known for low levels of intra and intergenerational income inequality (in particular, Nordic countries: Denmark, Finland, Iceland, Norway and Sweden). Over the past 20 years, these countries have SES gradients in math, reading and science scores, similar to other countries in Europe (especially in the most recent period). The reason this is surprising is that Nordic countries are known for having public services which are especially protective of poor families and children, through affordable daycare, extensive social safety nets, excellent universal health care, and high-quality public schools (Landerso and Heckman, 2017; Heckman and Landerso, 2022).

We also examine whether these patterns have persisted for decades or are a more recent phenomenon, using data from the Programme for the International Assessment of Adult Competencies (PIAAC), which measures skills in adulthood (Gal and Tout, 2014). For older cohorts in PIAAC (those born between 1957 and 1966), the cross-country correlation between SES gradients in skills and measures of intergenerational elasticity of earnings (IGE) from Corak (2013) is 0.85, reflecting the fact that both the IGE and SES gradients in test scores are relatively low in Nordic countries. In contrast, this pattern does not hold for more recent cohorts in PIAAC (born in 1987 or later, corresponding to those assessed by PISA), where this correlation drops to 0.13. Nordic countries no longer exhibit particularly high or low SES gradients in test scores, which is consistent with patterns observed in PISA data. This suggests that current IGE estimates primarily reflect social mobility among older cohorts born in the 1960s. Future IGE

¹Neal and Johnson (1996) and Carneiro et al. (2005) also document that test scores account for the bulk of racial gaps in earnings, while Cameron and Heckman (2001) show that they can fully explain racial gaps in schooling in the United States.

measures may align more closely with SES gradients observed in PISA scores, as individuals who took PISA assessments are only now entering the labor market.²

This shift in patterns raises important questions about their underlying factors, though it remains difficult to determine with certainty what drives these changes across cohorts. One possibility is that generous welfare systems reduce inequality and increase mobility in the short term, helping children from poor families overcome disadvantage. Particularly, those in the earliest cohorts we analyze benefited most from post-war expansions of welfare programs. But in the long term, as the welfare system becomes a structural feature of the economy, it is also possible that this does not provide strong incentives for poor children and their parents to invest in human capital to escape poverty.³ Instead, it may inadvertently incentivize rich parents to invest even more to maintain or enhance their relative advantage. Then the cross-country variation in SES gradients in learning may not reflect so much the equalizing forces of public investments, but rather the unequalizing differences in parental investments between poor and rich parents (Landerso and Heckman, 2017; Heckman and Landerso, 2022; Doepke and Zilibotti, 2019; Goldthorpe, 2016).

In support of this idea, we show that SES gradients in the quality of the home learning environment – henceforth referred to as parental investments and measured by the child’s access to adequate study conditions, technology, books and art – are as high in Nordic countries as elsewhere in Europe. In any given year, the cross-country correlation between SES gradients in learning and SES gradients in parental investments is high, ranging from 0.70 to 0.80.

Furthermore, we construct a panel of countries because PISA has been administered every 3 years for more than 20 years. Using this panel, we show that there is not only a strong cross-sectional correlation between SES gradients in parental investments and SES gradients in learning, but also that (within country) changes in SES gradients in parental investment are strongly related to (within country) changes in SES gradients in learning. Over 50% of the (within country) changes in SES gradients in test scores are explained by (within country) changes in SES gradients in parental investments, suggesting that the cross-sectional correlation is not solely due to time-invariant country effects. None of the other country characteristics we examine is as strong a predictor of SES gradients in learning, either in the cross-sectional or panel analysis.⁴

Our contribution is threefold, as discussed in detail in the paragraphs that follow. We show

²Some evidence is emerging that the IGE is rising in Nordic countries relative to the IGE elsewhere (Brandén et al., 2024; Ahrsjö et al., 2023).

³This argument bears some resemblance to the controversial gender equality paradox (e.g., Charles and Bradley (2009), Stoet and Geary (2018), Falk and Hermle (2018), Klinowski and Niederle (2025)), but applied to the relationship between socio-economic inequality and the poor’s incentives to learn.

⁴We consider variables that are plausible predictors of intergenerational transmission and for which we were able to assemble at least a partial panel between 2003 and 2018 (e.g., GDP per capita, returns to schooling, difference between top and bottom rates of income tax, government expenditure in education, and school segregation). We have experimented with other similar variables with no change in our main result.

how SES gradients in learning vary across countries and that they are primarily associated with SES gradients in parental investments, rather than with standard measures of inter- and intra-income inequality. This relationship is not only cross-sectional, but also holds when we examine changes within countries over time. Our findings suggest that education and anti-poverty policies may have limited impacts on SES gradients in learning because their equalizing effects may be undone by the stronger influence of unequal parental investments.

First, we extend the existing literature on intergenerational transmission of inequalities by broadening the focus beyond the transmission of income and years of schooling to learning outcomes, which reflect children’s human capital formation (e.g., [Black and Devereux \(2011\)](#)). This broader perspective acknowledges that the transmission of inequalities begins early in life and is shaped by multiple dimensions of development ([Carneiro and Heckman, 2003](#); [Cunha and Heckman, 2007](#)). Specifically, we document how the geography of SES gradients in test scores, aspirations, and non-cognitive skills has evolved over time, recognizing their multi-dimensional nature ([Heckman et al., 2006](#); [Deming, 2023](#)). This builds on a rich literature that has examined the geography of intergenerational transmission primarily through income and years of schooling. For example, [Chetty et al. \(2014\)](#) popularized this analysis by documenting the geography of the intergenerational transmission of income in the United States for the 1980-1982 birth cohort, while [Hertz et al. \(2008\)](#) and [Narayan et al. \(2018\)](#) document the intergenerational transmission of years of schooling across countries and over time.⁵

Second, we contribute to the literature examining cross-country differences in inequality transmission, which requires using similar datasets and methodologies across countries. Typical estimates of intergenerational income transmission can raise concerns about comparability, as they rely on different datasets, cohorts, and definitions of socio-economic status ([Mogstad and Torsvik, 2023](#); [Mello, Nybom, and Stuhler, 2025](#)). For example, [Haider and Solon \(2006\)](#) and [Grawe \(2006\)](#) show that estimates of intergenerational transmission of income across countries are sensitive to estimation methodology, lifecycle bias, and dataset differences.

To address these challenges and ensure comparability, on the one hand, we use data from PISA and PIAAC, which are specifically designed to allow for cross-country and longitudinal comparisons of children’s learning outcomes ([OECD, 2013](#); [OECD, 2023](#)). On the other hand, we emphasize the importance of selecting a consistent measure of socio-economic status (SES) to ensure comparability in parental outcomes. Income, while the most direct indicator, is unfortunately not available in PISA. We therefore explore widely used proxies of SES, such as

⁵[Blanden \(2013\)](#), [Corak \(2013\)](#), and [Stuhler \(2018\)](#) provide estimates of cross-country differences in intergenerational transmission of income, and others have explored regional mobility patterns in Africa, Latin America, Australia, Italy, England, Canada, and the United States ([Alesina et al., 2021](#); [Neidhofer et al., 2018](#); [Deutscher and Mazumder, 2020](#); [Acciari et al., 2022](#); [Bell et al., 2023](#); [Corak, 2020](#); [Card et al., 2022](#); [Hanushek et al., 2022](#)). Related work has also expanded beyond income and education, examining intergenerational transmission in wealth ([Charles and Hurst, 2003](#)), health ([Halliday et al., 2020](#); [Bencsik et al., 2023](#)), attitudes ([Dohmen et al., 2011](#)), and non-cognitive development ([Attanasio et al., 2025](#)).

parental education and the Economic, Social, and Cultural Status (ESCS) index developed by PISA. We recognize that each measure has strengths and limitations – parental education may vary across cohorts, altering the definition of high SES, while ESCS, though broader, lacks a natural scale and partially incorporates parental education.

Given these trade-offs, we primarily define SES gradients in learning by comparing math test scores between students whose mothers have at least completed high school and those who did not. This measure brings attention to a form of inequality transmission related to escaping the lower ranks of the education distribution, an issue we may be particularly concerned about from a policy perspective (Alesina et al., 2018). Importantly, we show that our results are robust regardless of which measure of SES we use. They are not driven by changes in the composition of mothers' education or the migrant population, and they remain consistent when excluding one country at a time or using alternative measures of the SES gradient.⁶ We finally focus on European countries rather than a global sample to examine nations with more similar institutional frameworks.

Third, we present new evidence of a strong panel relationship between changes in SES gradients in parental investment and learning across countries and over time. This comparative perspective reveals common patterns in the intergenerational transmission of inequality and underscores the persistent influence of the family. Notably, it helps explain why SES gradients in learning remain pronounced even in countries known for implementing equalizing family policies, such as the Nordic countries (Landerso and Heckman, 2017; Heckman and Landerso, 2022; Goldthorpe, 2016). Relative to the official PISA reports (see, for example, OECD (2010, 2014, 2018)), we combine data from all available periods, link the intergenerational transmission of learning to measures of inequality and its potential determinants, trace its evolution over time, and construct measures of SES in parental investment.

Using the PIAAC data, we also document a new pattern suggesting that the relationship between intergenerational and intra-generational inequality – commonly known as the Great Gatsby Curve (Corak, 2013) – has weakened over time. We show that younger PIAAC cohorts – those who took the PISA in 2003 – exhibit no correlation between SES gradients in test scores and either inequality or the IGE, while older cohorts born in the 1960s show strong correlations with both.⁷ These findings extend the work of Blanden et al. (2023), who previously reported no evidence of the Great Gatsby Curve when using PISA scores, although they did find such a relationship when using years of schooling (see also Doepke and Zilibotti (2017, 2019)). We

⁶Results are robust to other SES measures, such as using father's education, an indicator for whether mothers completed higher education, an indicator for whether mothers have at least the median education within the country, and the rank of the ESCS index developed by PISA. The ESCS combines information on parental education, occupation, and possessions (like owning a car, among other variables).

⁷Unfortunately, data on SES gradients in parental investments across countries for earlier PIAAC cohorts are unavailable. It is possible that high-SES parents were less protective in the past, when inequality levels were lower across European countries.

emphasize that PISA cohorts are younger than those typically used to study the Great Gatsby Curve in years of schooling. This distinction is important for understanding generational differences in inequality transmission.⁸

This paper is organized as follows. Section 2 discusses the data and the measures of intergenerational transmission of learning outcomes (i.e., SES gradients in test scores). Section 3 describes the geography and trends in the intergenerational transmission of learning in Europe over the last two decades. In Section 4, we compare the intergenerational transmission of learning and some of its possible determinants, exploring the role of parental investments. Section 5 concludes.

2 Intergenerational transmission of learning

2.1 PISA data

We use individual-level data from the PISA tests, administered every 3 years from 2000 to 2022 by the Organisation for Economic Co-operation and Development (OECD). These data contain information on 15-year-old students' learning outcomes in reading, mathematics and science. Learning outcomes are measured through standardized tests designed to assess students' preparedness in key subjects and for their adult life. Our analysis focuses on test scores for 24 European nations that have taken the tests for at least six waves in a row. Appendix Table A1 reports the countries in the analysis.

For the PISA test, typically, each country selects between 4,500 and 10,000 students through a two-stage stratified sampling technique. A random sample of at least 150 schools, enrolling 15-year-old students, is drawn first. Then, 35 students within each school are randomly selected to take part in the test. Each student responds to a randomly selected subset of questions from a larger pool.

PISA test scores are explicitly designed to allow comparisons across countries and over time (OECD, 2013; OECD, 2023). To make test scores comparable, they are not presented as point estimates, but as "plausible values". These values are computed by drawing random plausible values for each student from a probability distribution of test scores based on the student's answers (OECD, 2011).

To appropriately handle the structure of the PISA data, our analysis uses both plausible values and survey weights. We estimate the regression models iteratively, meaning that we re-run

⁸Durlauf et al. (2022) review several explanations for the Great Gatsby Curve, including credit constraints (Becker and Tomes, 1979; Becker, 1991; Solon, 1999), sorting and segregation (Durlauf and Seshadri, 2018), and aspirations (La Ferrara, 2019). Recent critiques, such as Mogstad and Torsvik (2023), argue that the curve is driven by a few outlier countries, while Karlson and Landersø (2021) suggest that intergenerational mobility reflects stages of development. Additionally, DiPrete (2020) notes that discrepancies arise from how inequality and mobility are measured, particularly regarding the role of taxes and transfers.

the same model multiple times – once for each plausible value provided for the dependent variable (e.g., students’ test scores, or ranks computed separately for each plausible value). In each iteration, a different plausible value is treated as the observed outcome variable, using survey weights to account for the complex sampling design of PISA (see, for example, [Jerrim et al. \(2017\)](#)). After estimating the model on each plausible value, we combine the resulting parameter estimates following Rubin’s rules ([Rubin, 1987](#)). This approach addresses the complex nature of PISA data in two key ways. First, using plausible values captures the latent nature of test scores and better reflects the uncertainty associated with them. Second, applying the provided survey weights enables estimation of population-level parameters, ensuring that findings are representative of the target population.

Every wave of the PISA data also includes a student, parent and school questionnaire. These questionnaires provide a wide range of information on students’ and parents’ characteristics, such as parents’ education, educational investment in their children, and home resources. We use this information to measure parents’ socio-economic status, which we discuss in detail in [Section 2.2](#).

We finally check to what extent the data on parental SES in PISA conform with data available from other sources. [Appendix Figure A1](#) presents a validation of the measures collected in the PISA dataset against external data sources from the World Bank Database. [Panel A of Appendix Figure A1](#) demonstrates a strong and close to 0.90 correlation between the proportion of mothers with at least upper secondary education (average over 7 PISA waves) and the proportion of females (aged 25+) with at least upper secondary education (World Bank database). Similarly, [Panel B of Appendix Figure A1](#) presents the correlation between the standard deviation of the PISA index of economic, social and cultural status (ESCS) and the Gini coefficient from the World Bank, which is also reasonably high, reaching almost 0.60.⁹

2.2 Measuring intergenerational transmission of learning

In this section, we discuss the measures of intergenerational transmission of learning (i.e., SES gradients in test scores/intergenerational mobility in learning) that we use throughout the paper. Each measure has its own advantages and disadvantages. To aid in interpretation, we begin with the association between the child’s test score and the mother’s education, using an indicator for whether the mother has completed at least upper secondary schooling ([Appendix Table A2](#) describes the ISCED levels).

For each student i in country c , cohort t and subject k , we estimate:

⁹The Educational, Social, and Cultural Status (ESCS) index in PISA is a composite measure that captures the socio-economic background of students participating in the assessment. It combines information on various indicators, such as parental education, occupation, and possessions (like owning a car, among other variables), to provide a summary measure of a student’s social and cultural capital.

$$R(Y_{itc}^k) = \alpha_{tc}^k + \beta_{tc}^k HS_{itc}^P + \epsilon_{itc}^k \quad (1)$$

where $R(Y_{itc}^k)$ is the test score percentile rank, computed at the European level, for subject k , for student i , in cohort t , living in country c . HS_{itc}^P is an indicator variable taking value 1 if student i 's mother has completed at least upper secondary education, and 0 otherwise. Throughout the analysis, we use the survey weights and plausible values provided by the OECD to account for the complexities of the PISA test and assure comparability of test scores across cohorts and countries (see Section 2.1).

β_{tc}^k measures the association between the mother's education (at least upper secondary education) and her child's rank in the distribution of math scores, for cohort t in country c . This represents a measure of the *SES gradient in learning*, which captures the difference in learning ranks between children whose mothers have different levels of schooling. Therefore, higher values of this parameter correspond to lower mobility (more persistence in the transmission of learning outcomes).

α_{tc}^k is the average learning rank for students whose mothers do not have a high school degree for cohort t in country c . This is a measure of *upward mobility* (higher values correspond to more mobility).

Both are important measures of intergenerational transmission of learning: the former is a measure of relative inequality between children of more and less advantaged families, while the latter is a measure of how well children of disadvantaged families perform.

In the paper, we present estimates for the math percentile rank, computed at the European level, and mother's education as we consider it easier to interpret and more comparable from a policy perspective. This also brings attention to a form of inequality transmission related to escaping the lower ranks of the education distribution, an issue we may be particularly concerned about (Alesina et al., 2018).

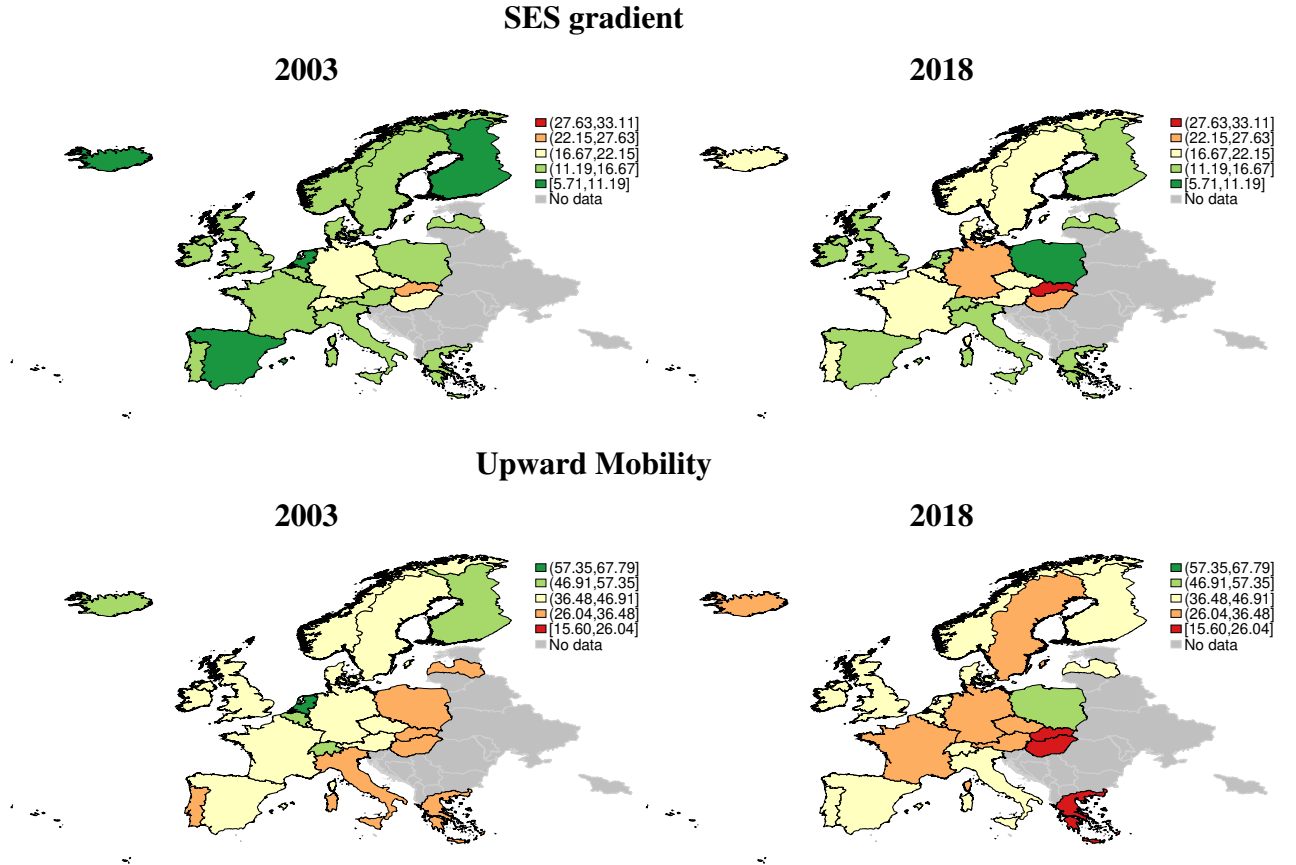
Importantly, we show that the estimates are robust to: (i) alternate measures of rank (European vs. country rank), (ii) definitions of parental socio-economic background, (iii) changes in composition and (iv) other measures of children's success. This is documented in Sections 3 and 4.3. For example, Section 4.3 also shows that the relationship between SES gradients in learning and SES gradients in parental investment is robust to excluding one country at a time, changes in the composition of mothers' education and migrants' population, and using alternative SES gradient measures detailed in the appendix.

Estimates of intergenerational transmission of learning are presented from 2003 because PISA begins to collect data on parental investment only from 2003 onwards. Measures of intergenerational transmission of learning for 2000 are available in Appendix Table A.2. We also present estimates for the 2022 wave (after COVID-19) in Section 4.6.

3 Intergenerational transmission of learning across Europe and over time

As discussed in the Introduction and Section 2, we begin by presenting the estimates for the intergenerational transmission using the student i 's math test score (rank), $R(Y_{itc}^{math})$, and start with the estimates of SES gradient (β_{tc}^{math}) and upward mobility (α_{tc}^{math}) for different European countries and different cohorts, based on equation (1). Figure 1 presents a heat map of these measures for 2003 and 2018 PISA waves. Green areas represent countries where inequality is less likely to be transmitted from parents to children (low β_{tc}^{math} in the case of the SES gradient in learning, and high α_{tc}^{math} in the case of upward mobility), whereas red areas indicate countries with greater transmission.

Figure 1: Heat map of intergenerational transmission of learning (math score)



Note. The heat maps present the estimates of intergenerational transmission of learning for each country and cohort (the estimates are based on equation (1), regression of the rank math score on dummy equal to 1 if the mother has at least upper secondary). The SES gradient in learning measure is β_{tc}^{math} from equation (1). The upward mobility (α_{tc}^{math}) is the average rank of test score of children whose mothers do not have upper secondary. The rank is computed at the European level. Green areas are the most mobile and red areas are the least mobile. The intervals are the same for each year.

Take for example Germany, which is one of the countries with the highest SES gradients in learning in our sample (even though it is not a country with particularly high or low IGE, as documented in Corak (2013)). In 2003, the SES gradient in learning, β_{tc}^{math} , was equal to

22, while the corresponding value for 2018 was 26. This means that, in Germany in 2003, children whose mothers completed upper secondary education were 22 percentiles above children whose mothers did not complete upper secondary education in the European distribution of PISA scores in math. This difference rose to 26 percentiles in 2018, which means that SES gradients in learning increased in Germany during this period. In 2003, the upward mobility, α_{tc}^{math} , was equal to 39, while the corresponding value for 2018 was 31. On average, children whose mothers did not complete upper secondary education scored in 39th percentile of the PISA distribution in 2003, while in 2018 their performance decreased to the 31st percentile (see Appendix Table A.2).

Together with Hungary and Slovakia, Germany is one of the three countries with the larger SES gradient in learning in Europe in 2003. Even though low-SES German students perform much worse than their high-SES counterparts, they perform much better than low-SES students in several other countries, such as Hungary, Slovakia, Poland, Italy, Greece, or Portugal. The countries with the lowest SES gradients in learning in 2003 are the Netherlands, Iceland, Spain and Finland. The Netherlands and Finland are also the two countries where students from low-SES backgrounds perform best across Europe.

By 2018, in most (but not all) countries, the link between a student’s learning and their parents’ SES became stronger compared to 2003. The countries with the largest SES gradients in learning are still Germany, Hungary and Slovakia, and as in 2003, students of low-SES status perform better in Germany than in either Hungary and Slovakia (which, together with Greece, are the countries with the lowest performance of low-SES students in 2018). In 2018, Poland shows higher levels of social mobility compared to the other countries in the sample.

There is however substantial stability between 2003 and 2018 in the relative position of countries in the sample. Appendix Figure A2 shows that the correlation in the estimates of the SES gradient in learning across countries between 2003 and 2018 is 0.7 (it is 0.8 between 2003 and 2009, and 0.93 between 2009 and 2018), while for upward mobility this correlation between 2003 and 2018 is 0.51 (0.65 for 2003 to 2009, and 0.94 for 2009 to 2018).

It is interesting that some of the larger deviations from this stability occur in Nordic countries. Sweden, Finland and Iceland (together with Slovakia and the Netherlands) are among the 5 countries where SES gradients in learning increased the most between 2003 and 2018 (Appendix Figure A3). In addition, we observe that SES gradients in test scores are similar in the United States (US) and Europe (Appendix Figure A4). Consistent with the arguments in Landerso and Heckman (2017) and Heckman and Landerso (2022), there is little difference between SES gradients in learning between the US and Denmark.

We also document that these patterns are robust to: alternate measures of rank, parental socio-economic background, and children’s success, as well as changes in the composition of the migrant population and mother’s education.

First, Appendix Table A4 shows that our results are robust to alternate measures of rank.¹⁰ This is because there is a good overlap in the test score distributions across different countries (Appendix Tables A5 and A6). The correlation between the two measures (rank at the European and rank at the country level) is high: 0.99 for β_{tc}^k and 0.67 for α_{tc}^k . Results are also robust to using alternate definitions of parental socio-economic background, such as: (i) an indicator for whether the father has completed upper secondary education, (ii) an indicator for whether the mother has higher education, (iii) an indicator for whether the mother has at least the median education within the country and (iv) the rank of the index of economic, social and cultural status (ESCS) constructed by the PISA. Appendix Table A4 shows the correlations among these estimates, which are usually above 0.60. Additionally, Section 4.5 presents SES gradients in other dimensions of children’s success, such as reading, science, educational aspiration, occupational aspiration and being on time to school.

Second, we document that these patterns are robust to considering only the native population (Appendix Figure A5). The correlation between the main measure and the one computed restricting the sample to the native population is 0.91, statistically significant. In addition, there is no correlation between the proportion of first- and second-generation migrant students in the country and our estimates of intergenerational transmission of learning (Appendix Table A7).

Third, we acknowledge that the proportion of children whose mothers have less than upper secondary schooling varies substantially across countries, and even across time. We document that there is no clear correlation between the proportion of mothers with low levels of education and our estimates of intergenerational transmission of learning (Appendix Table A8). If anything, there is a non-significant correlation which is hard to make sense under the most-standard selection mechanism. As the proportion of disadvantaged children declines, we would expect that the remaining disadvantaged children have worse unobservables. After all, these are the ones that stay behind in a society where disadvantage (at least on this measure) is declining. If that is a reasonable hypothesis, the estimated correlation with the proportion of disadvantaged children should go in the opposite directions for the SES gradient in learning and upward mobility.

Fourth, Section 4 further investigates the robustness of these results and documents that the relationship between SES gradients in learning and SES gradients in parental investment is robust to excluding one country at a time, compositional changes, and other measures of SES gradients in learning described in the appendix.

¹⁰The European rank is constructed by ordering each plausible value of the test scores in ascending order within Europe, using the weights provided by PISA. The advantage of using a European rather than a country-level rank is that the effect of mother’s education on test score is more comparable, as we are using the same distribution of test scores.

4 Correlates: intergenerational transmission of learning

What drives differences across countries, and changes over time, in SES gradients in learning? This is an important, but difficult question to answer. Even if we identify important correlates of cross-country variation in SES gradients in learning, it is difficult to establish that they are causal drivers.

It turns out, however, that it is difficult to identify important correlates of SES gradients in learning to start with. Surprisingly, there are not many variables strongly correlated with SES gradients in learning. We explore this for SES gradients in learning in Sections 4.1-4.3 and then turn to upward mobility and SES gradients in other traits in Sections 4.4 and 4.5.

4.1 Intergenerational transmission of learning and inequality

We begin by asking if there is a relationship between SES gradients in learning and inter and intra-generational inequality. There are several reasons why one might observe such a relationship. To the extent that human capital is an important determinant of earnings, it is perhaps natural to ask whether there is a relationship between SES gradients in learning and inequality across countries.

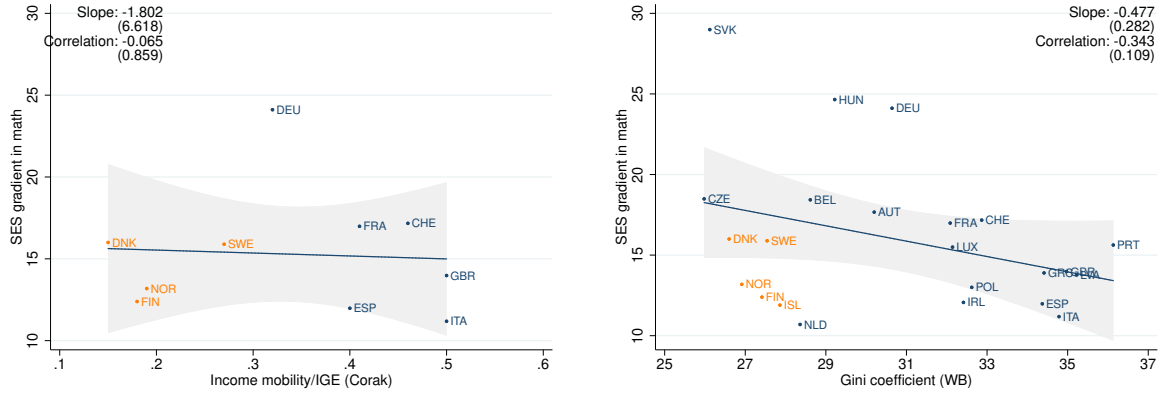
First, we document that SES gradients in learning are not correlated with estimates of the intergenerational mobility in income across countries (left panel of Figure 2). The estimates of income mobility come from Corak (2013), who compiles a dataset of IGE estimates from published studies, adjusting them for methodological comparability. These estimates are for the cohort of children born during the early to mid-1960s, with their adult outcomes measured in the mid to late 1990s (Corak, 2013). Since the IGE estimates from Corak (2013) are available for only a limited number of countries, we conduct a robustness check by reproducing the left panel of Figure 2 using estimates of inequality of opportunity, which cover a broader set of countries. The results are similar in Appendix Figure B1, corroborating our findings.

Second, we ask if there is a positive cross-country relationship between SES gradients in learning and inequality, known as the Great Gatsby curve (Corak, 2013), which has been widely discussed in the academic and popular press and documented both across and within countries (Chetty et al., 2014; Corak, 2020). It has however also been a subject of controversy (Mogstad and Torsvik, 2023). We document that SES gradients in learning are, if anything, negatively correlated with income inequality (right panel of Figure 2). As a whole, Nordic countries, which have the lowest income inequality in Europe, do not have smaller SES gradients in learning than countries with high income inequality, such as Great Britain, Poland, Latvia or Portugal.¹¹

We show that even when we use other measures of inequality or intergenerational transmis-

¹¹Nordic countries (Denmark, Finland, Iceland, Norway, and Sweden) are abbreviated with three-letter codes (DNK, FIN, ISL, NOR, and SWE) in the figures.

Figure 2: Correlation between SES gradients in math, IGE (Corak, 2013), and Gini



Note. The Left Figure presents the correlation between SES gradients in math (math - averaged over PISA years) and income mobility measure from Corak (2013). Corak (2013)'s estimates of income mobility are measured as the elasticity between paternal earnings and a son's adult earnings, using data on a cohort of children born during the early to mid 1960s and measuring their adult outcomes in the mid to late 1990s. These IGE estimates are derived from published studies, adjusted for methodological comparability in Corak (2013). The right figure presents the correlation between the SES gradient in math and Gini coefficient from the World Bank (averaged over all PISA years). The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

sion of learning, this correlation is often zero, but never positive (Blanden et al., 2023). We find no evidence for the Great Gatsby curve when we reproduce the right panel of Figure 2 by using: (i) SES gradients in learning from each PISA wave (Appendix Figure B2), (ii) the standard deviation of the ESCS index, built by PISA, to measure inequality in ESCS in any of the PISA cohorts (Appendix Figure B3), (iii) other definitions of SES gradients to measure inter-generational transmission of learning, such as rank-rank coefficients from regressing the rank of the test score on the rank of ESCS (Appendix Figure B4 and Appendix Table B1), and (iv) data from other international assessments, such as the Trends in International Mathematics and Science Study (TIMSS) and Progress in International Reading Literacy Study (PIRLS) studies, to measure learning (Appendix Figure B5 and Cardim and Carneiro (2021)).¹²

These findings suggest the possible emergence of a poverty trap in education – that is, a situation in which children from low-income families face persistent barriers to educational success, making it difficult for them to break out of the cycle of poverty across generations. Relative to other countries, rich parents in Nordic countries are not particularly impaired (e.g., by the progressivity of the tax system) when it comes to investing productively in their children. Conversely, poor parents in Nordic countries are not especially helped by the strong welfare state when it concerns their children's learning. One possible explanation, as proposed by Landerso and Heckman (2017) and Heckman and Landerso (2022), is that higher-SES parents are particularly adept at navigating and capitalizing on publicly provided benefits. As a result, the generous welfare provisions may inadvertently reinforce, rather than reduce, socio-economic

¹²Appendix Figure B6 shows a strong cross-country correlation between SES gradients in learning measured in the TIMSS/PIRLS and PISA.

disparities in education – an effect referred to as the Matthew Principle. We turn to this next by first exploring when these patterns have emerged in Section 4.2 and then by exploring other possible correlates of SES gradients in learning in Section 4.3.

4.2 Using PIAAC to study older cohorts

It is unclear whether the SES gradients in test scores observed today have persisted over multiple decades or are a more recent phenomenon. To investigate this question, we use data from the 2012 Programme for the International Assessment of Adult Competencies (PIAAC), which measures adult skills (Gal and Tout, 2014). This dataset enables cross-cohort comparisons and includes older cohorts for which we have measures of inequality (Appendix Figure B7).

First, we take advantage of a key feature of the PIAAC data: its youngest respondents (born in 1987 or later) belong to the same cohorts that participated in the 2003 PISA assessment. When comparing SES gradients in learning for these cohorts across the two surveys, we find a statistically significant correlation of 0.7 (Appendix Figure B8), indicating a strong consistency in SES-related learning outcomes captured by the two datasets.

Second, we investigate whether the relationship between SES gradients and inequality has changed over time. We divide the PIAAC sample into different cohorts and examine the cross-country correlation between SES gradients in test scores and two measures of inequality: (1) intergenerational earnings elasticity (IGE) for the 1960 cohort (Corak, 2013) and (2) inequality (Gini index from the World Bank, averaged over 1990-2010).

Figure 3 shows no clear correlation between SES gradients in test scores and IGE or inequality for younger cohorts, those also surveyed in early waves of PISA. In contrast, both correlations are strong and positive for older cohorts. Notably, for older cohorts – but not younger ones – SES gradients in learning are lowest in Nordic countries, which also have the lowest IGE and inequality levels.¹³

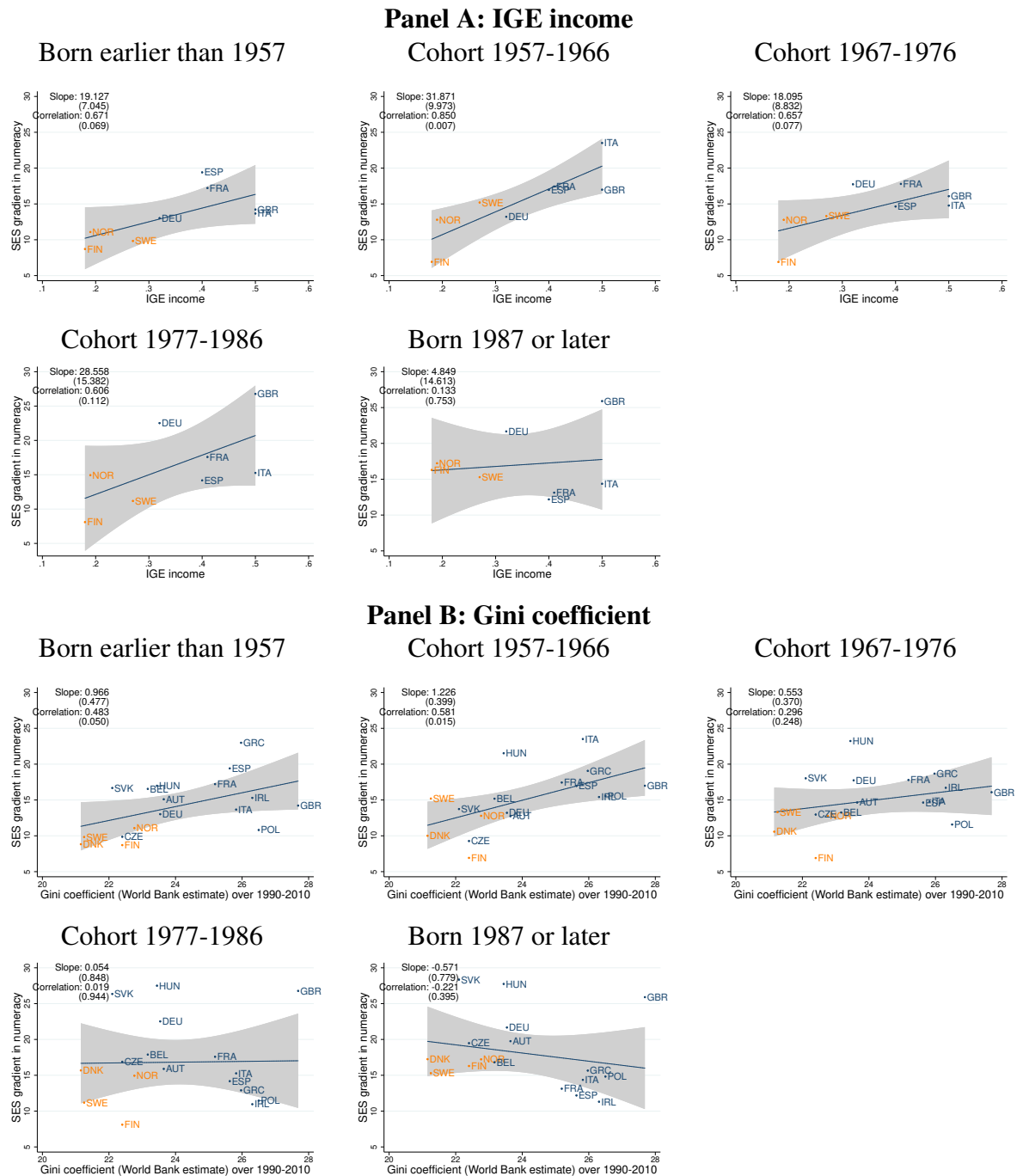
This finding offers new insights into how this relationship may have evolved over time, complementing the findings of Blanden et al. (2023), who reported no evidence of a positive relationship between SES gradients in learning and inequality in the cross section. The reason behind this finding could be that the PISA cohorts are younger than those typically used to study the Great Gatsby Curve in years of schooling.

Identifying a clear cause for the weakening of the Great Gatsby Curve is not straightforward. The evidence above could, for example, point to a shift in the underlying mechanisms – from broad-based parental investment responses to inequality towards dynamics concentrated in the tails of the SES distribution – suggesting that it may be important to account for potential nonlinearities and segment-specific incentives in theories linking inequality to mobility (Doepke

¹³While this could partly reflect age effects (i.e., performance differences based on test-taking age), it is more likely due to cohort effects, given the PIAAC test design (Hanushek et al., 2025).

Figure 3: PIAAC - correlation between SES gradients in numeracy, IGE (Corak, 2013), Gini coefficient

Figure 3: PIAAC - correlation between SES gradients in numeracy, IGE (Corak, 2013), Gini coefficient



Note. The figures present the correlation between SES gradients in numeracy and IGE income (Corak (2013)'s sample) in Panel A and the correlation between SES gradients in numeracy and Gini coefficient (averaged over 1990-2010) in Panel B. Corak (2013)'s estimates of income mobility are measured as the elasticity between paternal earnings and a son's adult earnings, using data on a cohort of children born during the early to mid 1960s and measuring their adult outcomes in the mid to late 1990s. These IGE estimates are derived from published studies, adjusted for methodological comparability in Corak (2013). The mobility measure is the β_{1c}^k from regressing the rank of numeracy skill on a dummy equal to 1 if the mother has at least upper secondary. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

4.3 Public and private investments in children

In this section, we look at potential correlates of SES gradients in learning. Since we have constructed a panel database of SES gradients in test scores across countries, it is possible to examine not only cross-sectional correlations, but also to study whether changes in particular variables are correlated with changes in SES gradients in test scores across different cohorts in each country.

Cross-sectional analysis We start by looking at cross-sectional associations. We consider different sets of variables that can contribute to differences in learning outcomes across SES groups: *socio-economic variables* (returns to secondary school from the OECD, and country GDP per capita from the World Bank database); *institutional variables* (the progressivity of the tax system, proxied by the difference between the top and lowest tax rates on personal income, and spending in public education from the World Bank database) and *education system variables* (age at which academic tracking starts from Eurydice and school-level segregation, which we measure from the PISA data as the R-squared from regressing the ESCS index on school fixed effects).

A final variable concerns parental investments in children, in particular, the SES gradients in parental investments. To measure parental investment in children, we estimate a factor model with categorical items using data from the students' questionnaire, which includes information on home educational resources, such as the child's access to adequate study conditions, technology, books and art. The factor model, described in detail in Appendix B.5.1, allows us to set the scaling assumption to obtain a comparable measure of parental investment across countries and over time.

We therefore measure the socio-economic gradient in parental investment by estimating a specification similar to equation (1), where we regress the rank of parental investment at the European level on an indicator variable equal to 1 if the mother has completed at least upper secondary. This measure of parental investment mainly reflects material investment, which has been shown to be particularly relevant for cognitive skills, while time investment tends to influence non-cognitive skills (Attanasio et al., 2020). We also discuss three plausible concerns about whether the SES gradients in parental investment capture the same information as the ESCS index built by PISA, a proxy of family socio-economic background.

First, the ESCS is measured by combining information about parental education, occupation and possessions, such as cars, while parental investment is measured by combining information on home environmental inputs, such as the possession of a desk, study space, and literature. Second, inequality in the ESCS and SES gradients in parental investments are not correlated, hinting that inequality in the ESCS and SES gradients in parental investment are measuring two different factors (Appendix Figure B9). Third, PISA also collects some information on parental

time investment from Portugal, Italy, Germany, and Belgium in 2012, 2015 and 2018. We find that, for this subset of countries, SES gradients in parental investment are positively correlated with SES gradients in time investment, with a correlation of 0.30 when pooling all three waves and 0.70 for 2018 alone.¹⁴

We begin by showing the correlation between each of these variables (averaged over all PISA years) and SES gradients in math scores (averaged over all PISA years) in Table 1. The variables most correlated with SES gradients in learning are: the age at which academic tracking starts, school segregation, and the SES gradients in parental investments. If we regress SES gradients in learning on all variables considered in Table 1, the school age tracking and SES gradients in parental investment survive and remain statistically significant (Column (8)).¹⁵

Table 1: Correlates of the SES gradient in learning

	SES gradient in math							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log GDP per capita	-1.901 (1.432)							-1.528 (1.379)
Returns to secondary school		0.285* (0.137)						0.012 (0.102)
Difference between top and lowest tax rates on personal income			-0.051 (0.072)					0.048 (0.053)
Government expenditure on education (% of GDP)				-0.498 (0.569)				0.979 (0.563)
School tracking age					-0.987*** (0.319)			-1.074** (0.456)
School segregation						38.118*** (10.741)		1.775 (15.890)
SES gradient in parental investment							0.841*** (0.128)	0.532** (0.202)
Observations	24	22	23	23	23	23	24	20
R ²	0.090	0.193	0.034	0.020	0.369	0.352	0.570	0.741

Note. The table shows the estimates from the SES gradient in math (averaged over all PISA years) and country-level variables (averaged over all PISA years). We consider different sets of variables. Some are socio-economic variables: the returns to secondary school from the OECD, country GDP. Some are institutional variables, such as the progressivity of the tax system, proxied by the difference between top and lowest tax rates on personal income, and spending in public education. Some are related to the education system, such as the age in which academic tracking starts, from Eurydice and school level segregation, which we measure from the PISA data as the R-squared from regressing the ESCS index on school fixed effects. The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Details on how the SES gradients in parental investment is computed are available in Appendix B.5.1. Robust standard errors (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

School tracking age has been documented to be relevant to explain the SES gradient in learning (Pekkarinen et al., 2009; Pekkala Kerr et al., 2013; Ferreira and Gignoux, 2013), a finding we confirm in our study. Systems where the placement of students into academic tracks starts early, such as in the case of Germany, would provide fewer chances for disadvantaged students to escape the least academic tracks, leading to larger SES gradients in test scores. In addition, a stronger sorting of students to schools based on SES (a higher level of SES

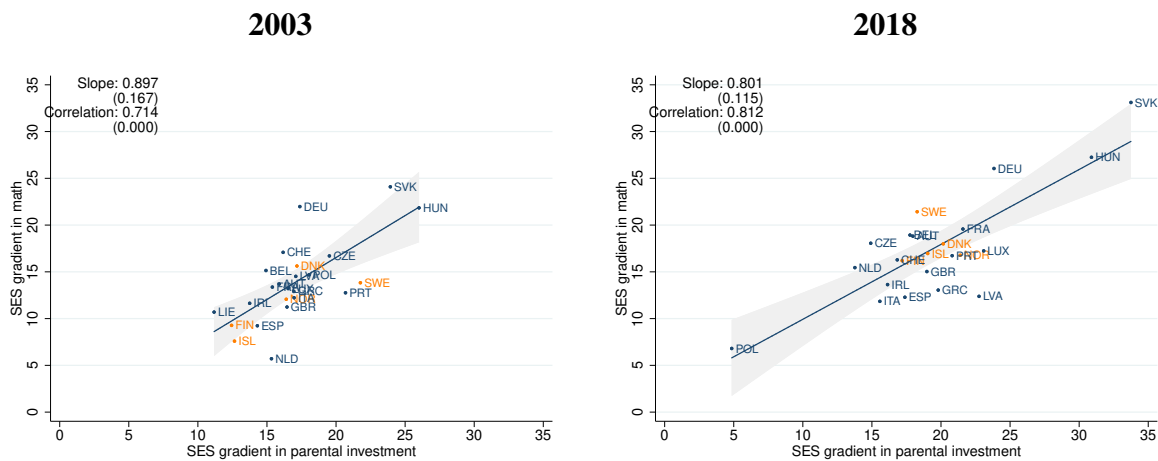
¹⁴We construct a measure for the SES gradients in time investment by combining information on frequency of the following activities between parents and children: parents discuss how well my child is doing at school, parents eat <the main meal> with my child around, parents spend time just talking to my child, parents help child with his/her reading and writing, parents discuss political or social issues with child, parents go to a bookstore or library with child, parents talk with child about what he/she is reading.

¹⁵Appendix Table B2 presents the standardized coefficients, with variables standardized to have a mean of 0 and a standard deviation of 1.

segregation in schools) could lead to lower social mobility, especially if high-SES schools are also better schools (because they are better equipped, have better teachers, or because students have higher achieving peers).

The fact that there is a strong relationship between SES gradients in learning and SES gradients in parental investments is also not surprising, especially if the parental investment index we construct is a powerful predictor of an individual's test scores.¹⁶ What is perhaps noteworthy is the strength of this association. Figure 4 plots SES gradients in learning and SES gradients in parental investment for two PISA cohorts: 2003 and 2018. The correlation between these two variables is 0.71 in 2003 and 0.81 in 2018, both of which are strikingly high values.

Figure 4: Correlation between SES gradients in learning and SES gradients in parental investment.



Note. The figures present the correlation between the SES gradient in learning (math) and SES gradients in parental investment. The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. Details on how the SES gradients in parental investment is computed are available in Appendix B.5.1. The rank is computed at the European level. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

Interestingly, Nordic countries, highlighted in orange in Figure 4, do not have particularly high or low SES gradients in parental investments. In 2003, Finland and Iceland had values for these gradients that were among the lowest in Europe, although it was not the case for Denmark, Norway, or Sweden. By 2018, however, their values were indistinguishable from those of a typical European country, even though the quantity and quality of public services available to disadvantaged children may be especially high in these countries. On top of that, more advantaged parents are still able to invest more in their children than less advantaged parents, and the degree to which they do so is as high in Nordic countries as anywhere else. Therefore, it is perhaps not surprising that SES gradients in learning are not especially low in

¹⁶We also compute bootstrapped standard errors with 50 iterations to account for uncertainty in the generated regressors (i.e., SES gradient for test scores and parental investment). Since the bootstrapped procedure produces standard errors similar to the robust method (bootstrapped SE: 0.042) and is computationally demanding, we compute bootstrapped standard errors only for the relationship between SES in test scores and parental investment.

Nordic countries.

Before turning to possible explanations for these cross-country differences, we first verify that our results are not driven by data peculiarities or model specification. We perform three robustness checks for the relationship between SES gradients in learning and SES gradients in parental investment. First, Appendix Table B3 shows that this relationship is robust to excluding one country at a time. Second, Appendix Table B4 shows that the association between the SES gradient in learning outcomes and the SES gradient in parental investment remains robust, when controlling for compositional factors, such as the percentage of migrant students and the percentage of mothers with less than a high school education. Third, we show that this relationship is robust to using the other measures of intergenerational transmission of learning described in the Appendix (Appendix Table B5). Specifically, Appendix Table B5 reproduces Columns (7) and (8) of Table 1 using the other measures of intergenerational transmission of learning.

Having established the robustness of our main findings, we next turn to potential mechanisms. These patterns suggest that the extent of public provision alone cannot fully account for cross-country differences in SES gradients in parental investment. To further investigate potential drivers behind these SES gradients in parental investment, at least in the cross-section, we explore two variables related to social norms and attitudes towards education: (i) confidence in the education system from the 2017 European Value Study (EVS), and (ii) perception that success depends strongly on parental wealth from the 2009 International Social Survey Programme (ISSP). The left panel of Appendix Figure B10 shows that a higher confidence in the education system is negatively correlated with larger socio-economic gradients in parental investment, while the right panel of Appendix Figure B10 shows that a stronger belief in the importance of a wealthy background is positively correlated with larger socio-economic gradients in parental investment. These findings align with Alesina et al. (2018) and Almås et al. (2020), who show that beliefs about social mobility and the fairness of inequality shape both preferences for redistribution and acceptance of inequality. In particular, our findings suggest that attitudes about the role of wealth and public institutions are also associated with private parental investment behaviors, potentially reinforcing socio-economic disparities.

Panel analysis The cross-sectional associations, reported in the previous section, are interesting but they are also difficult to interpret because they may be driven primarily by third factors. We examine if changes over time/across cohorts in at least some of these variables are correlated with changes over time/across cohorts in SES gradients in learning.

We therefore estimate the following model:

$$\beta_{tc}^{math} = X_{tc}\gamma + \tau_t + \theta_c + \epsilon_{tc} \quad (2)$$

where β_{tc}^{math} is the SES gradient in math for country c and cohort t , and X_{tc} is a country- and time-varying variable of interest. We include cohort fixed effects, τ_t , and country fixed effects, θ_c , to control for unobserved heterogeneity that is invariant across cohorts and countries. ϵ_{tc} is the residual, and we cluster standard errors at the country level.

The parameter of interest is γ and the estimates for all variables are presented in Table 2. We focus on the three main predictors of SES gradients in test scores: tracking age, school segregation, and SES gradients in investments. Because tracking age shows little to no variation over time during the period studied, we exclude it from this model and discuss the results for school segregation and SES gradients in investments.

Columns (5) and (6) of Table 2 document that changes in how segregated schools are across cohorts are not correlated with changes in SES gradients in learning, but changes in SES gradients in parental investments are strongly related to changes in SES gradients in learning.¹⁷ If we regress SES gradients in learning on all the country- and time-varying variables, considered in Table 2, the effect of SES gradients in parental investment survives and remains statistically significant with a similar magnitude (Column (7)).¹⁸

Table 2: Panel regressions: SES gradient in learning

	SES gradient in math						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log GDP per capita	-1.477 (4.119)						-5.149 (5.433)
Returns to secondary school		0.106* (0.053)					0.051 (0.086)
Difference between top and lowest tax rates on personal income			0.028 (0.066)				0.083 (0.110)
Government expenditure on education (% of GDP)				-0.342 (0.565)			-0.354 (0.563)
School segregation					7.840 (11.276)		2.549 (15.906)
SES gradient in parental investment						0.564*** (0.129)	0.631*** (0.189)
Observations	160	118	137	117	159	142	74
R^2	0.802	0.841	0.808	0.819	0.802	0.881	0.916
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

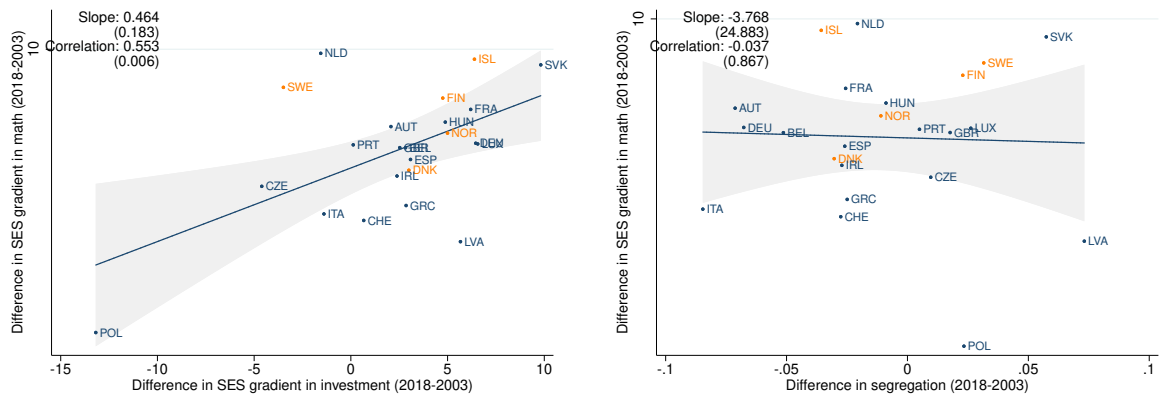
Note. The table presents the panel estimates of the SES gradient in learning on economic variables with year and country fixed effects (equation (2)). The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Clustered standard errors at the country level (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 5 shows graphically the relationship between SES gradients in learning and SES gradients in parental investment by plotting changes in SES gradients in learning between 2003 and 2018 against changes in SES gradients in investment (left panel) and changes in segregation (right panel) during the same period. some of the countries showing the largest increase in

¹⁷We also compute bootstrapped standard errors based on 50 repetitions to incorporate the uncertainty in the estimated SES gradient in test scores and parental investment. Importantly, the bootstrapped standard error (0.067) is similar to the cluster robust standard errors. Given the computational intensity of this process and the similarity of estimates standard errors, we bootstrap only the relationship between SES in learning and SES in parental investment.

¹⁸Appendix Table B6 presents the standardized coefficients, with variables standardized to have a mean of 0 and a standard deviation of 1.

Figure 5: Correlation: changes in SES gradients in learning, SES gradients in investment and segregation between 2003 and 2018.



Note. The figure presents the correlation between the changes in SES gradients in learning (math), changes in the socio-economic gradients in parental investment, and changes in segregation between 2003 and 2018. The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. Details on how the SES gradients in parental investment is computed are available in Appendix B.5.1. The rank is computed at the European level. Segregation is measured in the PISA data as the R-squared from regressing the ESCS index on school fixed effects. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

SES gradients in parental investments are the Nordic ones – Norway, Finland, Iceland, and Denmark. These results are, however, not driven by changes in inequality in these countries. We indeed show that there is no strong correlation between inequality in ESCS and SES gradients in parental investments (Appendix Figure B9). Nonetheless, the correlation between changes in SES gradients in parental investment and changes in SES gradients in learning is strong, at 0.55, suggesting that increased disparities in investment are associated with widening achievement gaps (Figure 5).

There could be several potential drivers of SES gradients in learning. For example, the provision of public standardized education services, and the degree of inequality in school quality between schools attended by poor and rich students could in theory play a big role. In practice, we do not find evidence that this is the case. Although we see that the degree to which students segregate into schools based on their SES is associated with SES gradients in learning in the cross-section, we do not find such a relationship when we look at the panel. We cannot rule out that whatever factors driving differences in SES gradients in parental investments across countries or time also have an independent effect on SES gradients in learning, but it is interesting to think about what these factors may be. In sum, our results suggest that the roots of inequality in child outcomes lie in the family, despite equalizing policies, such as those observed in Nordic countries, aimed at leveling the playing field.

Importantly, we do not claim to have estimated causal effects, which would require identifying sources of exogenous variation. Our results should be interpreted as descriptive. Rather, the purpose of the robustness checks we perform is to verify that the observed panel relationship

between SES gradients in learning and SES gradients in parental investment is not driven by specific country outliers (Appendix Table B7), by changes in sample composition (Appendix Table B8), or by the choice of measurement for intergenerational transmission of learning (Appendix Table B9), but instead reflects a stable association.

4.4 Upward mobility

There are many measures of social mobility, and one that has become prominent is upward mobility, which measures the average outcomes of individuals coming from a disadvantaged background (e.g., Chetty et al. (2014); Deutscher and Mazumder (2023)). In our setting, this corresponds to the average rank in the PISA score distribution for individuals whose mothers have less than upper secondary schooling. This measure is relevant when one is concerned about the outcomes of the most disadvantaged groups within a country and may vary across countries for different reasons from the ones we focused on so far. For example, a country with an excellent education system can make sure the poor have good learning levels, without changing the SES gradients in learning.

First, we document that upward mobility and average PISA scores are strongly, though not perfectly, correlated (Appendix Figure C1): these patterns are not just explained by average differences across countries.¹⁹ Second, and perhaps unsurprisingly, while country rankings for upward mobility and the SES gradient in learning are not identical, they are strongly correlated: 0.60 in 2003 and 0.80 in 2018 (Appendix Figure C2).

Finally, regarding the correlates of upward mobility, they work as the correlates of the SES gradient in learning. There is no correlation with inequality across countries or over time. In the cross section, school segregation, age at first tracking, and SES gradients in parental investments have the strongest correlations with upward mobility in learning. In the panel, once we control for country and cohort fixed effects, only changes in the SES gradients in parental investment remain an important predictor of changes in upward mobility within country and over time (Appendix Table C1).

4.5 SES gradients in other traits

PISA has also rich information on other child outcomes, which we can use to examine SES gradients in learning and traits more broadly. First, there are the well-known PISA scores, not only in math, but also in reading and science. Second, there are measures of occupational and educational aspirations for each child. Third, there are measures of self-efficacy and being on time to school. We describe these measures below and refer to Appendix C.3 for further details.

¹⁹For this reason, when we explore the correlates of the SES gradient in learning, we do not observe the same patterns as the ones noted by Woessmann (2016).

It is important, to the extent possible, to look beyond SES gradients in one academic subject, as human capital is multi-dimensional (see, for example, [Heckman et al. \(2006\)](#) and [Deming \(2023\)](#)). To measure students' educational and occupational aspirations, we follow [La Ferrara \(2019\)](#), defining occupational aspiration with a dummy equal to 1 for students, who expect to have a white collar job, and educational aspiration with a dummy equal to 1 for students, who expect to study at university. Self-efficacy captures the extent to which individuals believe in their own ability to engage in certain activities and perform specific tasks, especially when facing adverse circumstances ([Bandura, 1977, 1991](#)). To measure non-cognitive skills, we look at the probability of being on time to school ([Cunha and Heckman, 2007](#); [Borghans et al., 2008](#)).

We measure the SES gradient and upward mobility in these other traits by estimating equation (1), where the outcome is now one of the traits we describe above. Upward mobility is then the average trait for the children whose mothers have not completed upper secondary education. The SES gradient in the trait is instead the difference in a certain trait between children whose mothers have completed at least upper secondary and mothers that have not.

Appendix Tables [C2](#) and [C3](#) present the correlation of the measures of intergenerational transmission in other traits within a given year. As expected, it makes very little difference to measure SES gradients in math, reading or science. SES gradients in aspirations have strong positive correlations with SES gradients in test scores, but which are substantially away from 1, suggesting that these are different traits. Still positive, but less correlated with SES gradients in test scores, are SES gradients in being on time to school.²⁰

Importantly, we still uncover a strong relationship between SES gradients in these other traits and the SES gradients in parental investment, while we do not find a correlation between SES gradients in these other traits and inequality (Appendix Table [C4](#)). This reiterates the points above and provides additional evidence on the importance of family behavior for SES gradients in children's outcomes.

4.6 SES gradients in learning before and after COVID-19

We finally explore the recently-released PISA data from the 2022 wave and compare SES gradients in test scores in 2018 and 2022 (i.e., before and after the COVID-19 pandemic). Although it could be tempting to attribute any changes to students' experience during the pandemic, there are longer-term trends in these gradients that should be considered first. We focus on the SES gradients in test scores, while other studies have documented a decrease in the average scores and in the share of students who are able to reach a minimum competence (e.g., [European Commission \(2024\)](#)).

From the 2022 data, we still observe that SES gradients in test scores are as high for the

²⁰Similar patterns are also found for the SES gradients in self-efficacy, which is available for 2003 and 2012 (Appendix Figure [C3](#)).

Nordic countries as for the remainder of the countries in the sample (Appendix Figure D1). Appendix Figure D2 shows that there is a strong correlation between SES gradients in math scores in 2018 and 2022, and that the level of these gradients has not changed significantly between these two years. This can also be observed in Appendix Figure D3, presenting the trends in the SES gradient in learning and upward mobility across countries, with the 2022-PISA wave. Although it is true that there are several countries for which the SES gradient in PISA scores reached its highest value in 2022, there are also many others where this is not the case. More importantly, there is no obvious change in the evolution of these gradients over time. For most if not all countries, the 2022 data seem to follow a trend, rather than exhibiting a break from the past. Such patterns are also observed when we examine the evolution of SES gradients in parental investments between 2018 and 2022 (Appendix Figure D4).

We also continue to see a strong correlation between SES gradients in parental investment and SES gradients in test scores (Appendix Figure D5). In fact, comparing Appendix Figure D5 with Figure 4, we see that, if anything, the relationship between SES gradients in parental investment and SES gradients in test scores becomes stronger (correlation: 0.89). On the other hand, we find no correlation between school closures during COVID-19 and SES gradients in learning (Appendix Figure D6), with school closures potentially serving as a proxy for public investment. These results suggest that families still played an important role in the COVID-induced disruption in learning.

5 Conclusion

Standardized international tests show significant differences in student knowledge across countries, even within Europe. Using PISA, we document that there are also staggering differences in the magnitude of SES gradients in test scores across countries, in the test performance of the most disadvantaged students in each country, and how these vary over time. For example, in 2003, the gradients in PISA scores between children whose mothers have or have not completed at least upper secondary education are (to give just a few examples) 5 percentile ranks in the Netherlands, 11 in Great Britain, 16 in Denmark, and 22 in Germany. Fast forward to 2018, and these values are respectively, 16, 15, 18 and 26. A similar picture is found for upward mobility, where we observe a decline in the average performance of disadvantaged students.

These patterns reveal that changes over time are not uniform, and that some countries – even those with generous and universal social benefits – face persistent SES gradients in learning. Nordic countries, often regarded for their egalitarian welfare systems, do not exhibit particularly low SES gradients when compared to other European countries, especially in the most recent PISA cohorts. In fact, there is no correlation between the SES gradients in learning and inter/intra-generational income inequality, neither in the cross-section nor the time series.

PIAAC data suggest that this may not have been the case in the past.

Building on these observations, we next investigate the main correlates of SES gradients in learning. We find that these gradients are associated with the level of SES segregation in schools, the age at which tracking first starts in the education system, and the SES gradients in parental investments. Of these, only the latter shows a cross-sectional association that persists even when we consider the entire panel and account for country and cohort fixed effects.

These findings suggest that while social policies contribute to addressing SES gradients in learning, family behaviors still play an important role in perpetuating these gaps. If other countries were to import the same type of education and social policies as Nordic countries, it is not clear that they would experience substantial reductions in SES gradients in learning. Policies could also aim to open up bottlenecks along the path to success to break the intergenerational transmission of inequality ([Fishkin, 2014](#); [Durlauf et al., 2024](#)). Addressing such bottlenecks may require interventions that operate beyond conventional welfare and education systems.

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Appendices to "Parental Investments and Socio-Economic Gradients in Learning across European Countries"

A Descriptive evidence

Table A1: Mapping of three-letter country codes to full country names

Country Code	Full Country Name
AUT	Austria
BEL	Belgium
CHE	Switzerland
CZE	Czech Republic
DEU	Germany
DNK	Denmark
ESP	Spain
FIN	Finland
FRA	France
GBR	United Kingdom
GRC	Greece
HUN	Hungary
IRL	Ireland
ISL	Iceland
ITA	Italy
LIE	Liechtenstein
LUX	Luxembourg
LVA	Latvia
NLD	Netherlands
NOR	Norway
POL	Poland
PRT	Portugal
SVK	Slovakia
SWE	Sweden

A.1 Data validation of mother's education and ESCS index

Answers on parental education were coded by PISA into internationally-comparable educational qualifications, following the international Standard Classification of Education (ISCED), as shown in Appendix Table [A2](#).

Panel A of Appendix Figure [A1](#) presents the correlation between the proportion of mothers with at least upper secondary education (average over 7 PISA waves) and the proportion of females (aged 25+) with at least upper secondary education (World Bank database).

We note that the proportion of mothers with at least upper secondary education is higher than proportion of females (aged 25+) with at least upper secondary education from the World Bank. This is because in PISA we have mothers, while in the World Bank we have the female population. The correlation between the two is however high - close to 0.90.

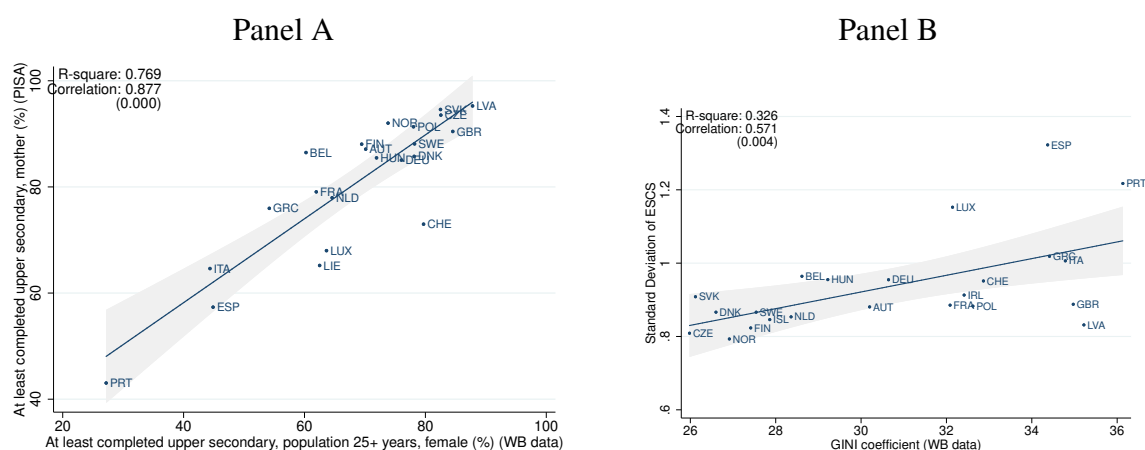
Table A2: International Standard Classification of Education (ISCED) levels

Level	ISCED
0	None
1	ISCED 1 (primary education)
2	ISCED 2 (lower secondary)
3	ISCED Level 3B or 3C (vocational/pre-vocational upper secondary)
4	ISCED 3A (upper secondary) and/or ISCED 4 (non-tertiary post-secondary)
5	ISCED 5B (vocational tertiary)
6	ISCED 5A, 6 (theoretically oriented tertiary and post-graduate)

Note. The table presents the International Standard Classification of Education (ISCED) classification. PISA codes the answers on parental education into internationally-comparable educational qualifications, following the international Standard Classification of Education (ISCED).

Panel B of Appendix Figure A1 presents the correlation between the standard deviation of the PISA index of economic, social and cultural status (ESCS) and the Gini coefficient from the World Bank.

Figure A1: Correlation between PISA and the World Bank data



Note. Panel A presents the correlation between the proportion of mother with at least upper secondary education in PISA and the proportion of females (aged 25+) with at least upper secondary education in the World Bank database. Panel B presents the correlation between the standard deviation of the ESCS index in PISA and the Gini coefficient from the World Bank. SES in Germany has been harmonized from 2012. In each panel, we compute each statistics at the country level for each wave and then link it to the respective wave of the World Bank database. The figures present the correlation between the averages computed over the 7 PISA waves. Confidence interval at 95% level in gray. The notes in the figures report the R-squared and the correlation between variables with the p-value in parenthesis.

A.2 Descriptive statistics on intergenerational transmission of learning

Table A3: Descriptive statistics on intergenerational transmission of learning

Country	Upward 2000	SES gap	Upward 2003	SES gap	Upward 2006	SES gap	Upward 2009	SES gap	Upward 2012	SES gap	Upward 2015	SES gap	Upward 2018	SES gap
AUT	49.17	10.53	41.04	13.69	38.60	17.58	32.25	19.95	34.43	20.64	31.07	22.45	34.65	18.82
BEL	44.90	20.18	49.75	15.14	44.27	18.47	42.71	17.31	39.07	20.08	38.10	18.89	37.84	18.93
CHE	50.35	18.68	47.77	17.08	50.50	15.50	49.05	16.91	47.13	17.40	44.05	18.32	43.11	16.28
CZE	35.89	16.19	41.33	16.70	31.39	25.14	31.39	18.56	32.71	18.99	35.77	15.78	35.53	18.06
DEU	30.11	25.76	38.68	21.98	38.47	20.77	41.39	20.28	30.74	29.18	33.13	24.76	30.90	26.05
DNK	43.94	16.67	43.01	15.62	44.61	14.00	39.11	15.99	36.80	16.81	43.30	14.90	37.71	17.99
ESP	39.56	14.11	41.81	9.24	40.82	11.31	39.94	11.57	38.57	12.81	39.55	12.52	37.18	12.28
FIN	57.59	7.53	56.37	9.28	58.85	9.51	51.42	13.85	44.44	13.99	39.91	16.39	38.42	16.19
FRA	49.58	12.12	45.65	13.37	40.21	16.34	37.43	18.99	35.01	19.56	34.80	18.94	33.48	19.58
GBR	50.32	13.91	44.25	11.24	41.80	11.55	35.88	16.07	37.80	14.50	36.54	15.55	39.25	15.03
GRC	29.12	15.22	26.23	12.91	29.40	14.91	30.76	13.35	25.46	14.20	26.03	13.62	25.11	13.05
HUN	29.80	22.15	29.23	21.84	30.12	22.57	25.66	27.37	23.74	24.51	21.46	26.87	21.38	27.25
IRL	46.91	9.89	42.81	11.63	42.76	12.98	39.37	11.37	42.27	11.02	40.44	13.90	38.83	13.64
ISL	51.71	9.64	50.45	7.59	45.30	12.68	43.51	13.59	40.71	10.92	38.24	11.87	35.58	16.96
ITA	34.50	9.13	33.42	12.23	34.44	10.76	39.79	10.23	38.47	11.81	40.16	12.28	38.76	11.84
LIE	52.11	14.64	55.19	10.70	51.23	13.43	48.41	20.14	49.79	14.83				
LUX	32.58	10.75	40.89	13.21	40.79	13.76	36.82	17.88	35.84	18.41	35.34	17.21	34.18	17.24
LVA	28.72	14.36	31.47	14.52	33.03	15.58	36.67	9.17	33.34	15.67	31.95	14.66	38.17	12.38
NLD	67.79	8.02	60.15	5.71	53.66	10.41	48.61	13.28	51.18	9.12	44.57	12.88	44.17	15.45
NOR	46.24	7.60	38.70	12.07	37.26	13.49	38.70	13.06	35.98	13.68	38.08	15.62	36.54	16.77
POL	38.71	6.80	33.65	14.68	34.57	17.49	32.94	18.26	36.98	20.24	47.33	6.73	50.01	6.80
PRT	35.28	11.88	35.92	12.76	36.80	14.89	39.72	18.57	40.18	17.68	40.94	16.86	39.87	16.73
SVK			27.84	24.09	24.14	27.49	22.74	28.40	15.60	32.11	17.85	28.76	16.36	33.11
SWE	47.35	9.50	42.66	13.83	41.81	13.38	35.18	17.42	32.87	14.30	31.55	21.36	33.69	21.43

Note. The table presents the estimates of the SES gradient and upward mobility. The estimates are based on equation (1), regression of the rank math score on dummy equal to 1 if mother has at least completed upper secondary. The rank is computed at the European level. Higher values of the SES gradient in learning correspond to lower mobility.

A.3 Robustness to alternate measures

Appendix Table A4 shows that the correlation between the alternate measures is relatively high, providing confidence that the main results are robust to different definitions of intergenerational transmission of learning.

Reading test score - we estimate the intergenerational transmission of learning (1) by using the rank of the reading test score.

Rank at the country level - we estimate the intergenerational transmission of learning (1) by using the rank of the test score at the country level. The country rank is constructed by ordering each plausible value of the test scores in ascending order within each country, using the weights provided by PISA. Thus, each student has one rank for each plausible value. By defining within-country ranks, the estimated social mobility parameter compares the positions in the country's outcome distribution of children at different points in the country's SES distribution. The disadvantage of this measure is the difficulty in comparing estimates across countries, since the distribution of test scores may vary considerably from country to country. We note that constructing a European rank may be problematic if the test score distributions do not overlap across countries. Appendix Tables A5 and A6, however, show a good overlap of the test score distribution across countries.

Alternate measures of parental socio-economic status.

Father's education - we compute the main measure of SES gradient in learning in math from equation (1) by using the father's education dummy, instead of the mother's ($\beta_{tc}^{math-father}$).

Mother's higher education - we define an indicator variable taking value 1 if i 's mother has completed higher education (HE_{itc}^P), and 0 otherwise (Appendix Table A2). For each student i , country c , cohort t and subject k , we estimate:

$$R(Y_{itc}^k) = \alpha_{tc}^{k-uni} + \beta_{tc}^{k-uni} HE_{itc}^P + \epsilon_{itc}^k \quad (3)$$

Rank of ESCS - we estimate a rank-rank regression, where we regress the rank of the math test score on the rank of the socio-economic and cultural status (ESCS), which can be considered as a proxy for permanent income. For each student i , country c , cohort t and subject k , we estimate:

$$R(Y_{itc}^k) = \mu_{tc}^k + \rho_{tc}^k R(Y_{itc}^P) + \epsilon_{itc}^k \quad (4)$$

where $R(Y_{itc}^k)$ is the test score (rank) for subject k for student i in cohort t living in country c and $R(Y_{itc}^P)$ is i 's the index of socio-economic and cultural status (rank). The index of socio-economic and cultural status (ESCS) is derived from three factors: highest parental occupation, highest parental education, and an Item Response Theory (IRT) scale based on student reports on home possessions. PISA has constructed the index in 2015, making it comparable over the waves. ESCS components and the ESCS model have changed over cycles and with that, ESCS scores are not comparable across cycles directly. To enable a trend study, PISA 2015 computes the ESCS measure for the 2015 cycle and also recomputes it for the earlier cycles using the same methodology.²

Above median education - we use a different definition of socio-economic status by defining the indicator variable taking value 1 if i 's mother has at least the median education within the country (ME_{itc}^P), and 0 otherwise. For each student i , country c , cohort t and subject k , we estimate:

$$R(Y_{itc}^k) = \alpha_{tc}^{k-med} + \beta_{tc}^{k-med} ME_{itc}^P + \epsilon_{itc}^k \quad (5)$$

where β_{tc}^{k-med} is the difference in math rank between children whose mothers have below and at least the median education within the country and α_{tc}^{k-med} is the children's average rank when the mother has below the median education within the country.

²Some examples of changes over the different waves are the following. The mapping of ISCED levels to years of schooling was updated in 2009 and 2015 for some countries, taking into account changes in countries' learning systems. Indicators of home possession (HOMEPOS) have been dropped or added in all PISA cycles (except in PISA 2012) taking into account social, technical and economic changes in participating societies (OECD, 2012).

Table A4: Correlation among alternate intergenerational transmission of learning measures

	Relative mobility						
	β_{tc}^{math}	$\beta_{tc}^{math-father}$	β_{tc}^{math-c}	β_{tc}^{read}	$\beta_{tc}^{math-uni}$	$\beta_{tc}^{math-med}$	ρ_{tc}^{math}
β_{tc}^{math}	1						
$\beta_{tc}^{math-father}$	0.942***	1					
β_{tc}^{math-c}	0.992***	0.928***	1				
β_{tc}^{read}	0.973***	0.941***	0.957***	1			
$\beta_{tc}^{math-uni}$	0.337	0.405*	0.342	0.292	1		
$\beta_{tc}^{math-med}$	0.714***	0.784***	0.701***	0.678***	0.739***	1	
ρ_{tc}^{math}	0.763***	0.753***	0.711***	0.772***	0.376*	0.661***	1

	Upward mobility						
	α_{tc}^{math}	$\alpha_{tc}^{math-father}$	α_{tc}^{math-c}	α_{tc}^{read}	$\alpha_{tc}^{math-uni}$	$\alpha_{tc}^{math-med}$	μ_{tc}^{math}
α_{tc}^{math}	1.000						
$\alpha_{tc}^{math-father}$	0.967***	1.000					
α_{tc}^{math-c}	0.670***	0.652***	1.000				
α_{tc}^{read}	0.888***	0.911***	0.774***	1.000			
$\alpha_{tc}^{math-uni}$	0.686***	0.621***	0.007	0.399*	1.000		
$\alpha_{tc}^{math-med}$	0.852***	0.866***	0.248	0.668***	0.861***	1.000	
μ_{tc}^{math}	0.737***	0.706***	0.362*	0.644***	0.625***	0.632***	1.000

Note. The table presents the correlation among different measures of intergenerational transmission of learning (averaged across PISA waves). β_{tc}^{math} is the main measure of the SES gradient in learning in math from equation (1), $\beta_{tc}^{math-father}$ is the main measure of the SES gradient in learning in math from equation (1) when using the father's education dummy instead of the mother's, β_{tc}^{read} is the main measure of the SES gradient in learning in reading from equation (1), β_{tc}^{math-c} is the SES gradient in learning in math from equation (1) when rank is computed within the country, $\beta_{tc}^{math-uni}$ is the SES gradient in learning in math from equation (3), $\beta_{tc}^{math-med}$ is the SES gradient in learning in math from equation (5), ρ_{tc}^{math} is the SES gradient in learning from equation (4) using the rank of the parents' ESCS.

α_{tc}^{math} is the main measure of upward mobility in math from equation (1), $\alpha_{tc}^{math-father}$ is the main measure of upward mobility in math from equation (1) when using the father's education dummy instead of the mother's, α_{tc}^{read} is the main measure of upward mobility in reading from equation (1), α_{tc}^{math-c} is upward mobility in math from equation (1) when rank is computed within the country, $\alpha_{tc}^{math-uni}$ is upward mobility in math from equation (3), $\alpha_{tc}^{math-med}$ is upward mobility in math from equation (5), μ_{tc}^{math} is upward mobility from equation (4) using the rank of the parents' ESCS. (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.4 Descriptive statistics on test scores

Table A5: Test score statistics, 2003

Country Name	Reading							Mathematics						
	Min	P25	P50	P75	Max	Mean	N	Min	P25	P50	P75	Max	Mean	N
AUT	16.62	423.47	498.69	565.05	786.68	490.69	4597	205.42	439.37	506.22	571.43	790.92	505.61	4597
BEL	31.54	439.93	521.61	587.04	792.17	506.99	8796	114.3	456.22	537.14	611.17	834.06	529.29	8796
CHE	96.51	438.52	505.98	565.28	787.34	499.12	8420	164.42	460.76	529.66	594.98	839.12	526.55	8420
CZE	160.17	427.78	494.53	555.26	800.9	488.54	6320	199.41	449.4	516.77	584.41	857.83	516.46	6320
DEU	69.23	418.51	503.62	571.75	782.91	491.36	4660	147.31	432.17	509.16	578.31	801.06	502.99	4660
DNK	86.67	437.75	498.47	553.38	760.64	492.32	4218	157.36	453.17	516.08	578.24	777.16	514.29	4218
ESP	7.39	420.5	486.99	548.19	756.96	480.54	10791	115.42	426.19	487.1	546.44	787.38	485.11	10791
FIN	152.82	493.63	549.2	598.78	803.6	543.46	5796	223.15	488.24	543.73	602.63	837.64	544.29	5796
FRA	119.61	436.03	504.99	565.43	757.01	496.19	4300	189.7	449.06	514.24	575.25	782.94	510.8	4300
GBR	106.88	445.47	511.87	573.58	809.24	507.01	9535	146.8	444.1	509.78	572.6	808.9	508.26	9535
GRC	-17.89	406.34	479.84	545.95	780.09	472.27	4627	77.85	382.36	446.11	507.87	768.66	444.91	4627
HUN	129.4	422.03	486.57	546.47	752.47	481.87	4765	153.86	426.09	489.97	555.88	794.66	490.01	4765
IRL	150.09	460.18	520.64	576.9	754.22	515.48	3880	203.47	444.99	503.47	561.88	789.53	502.84	3880
ISL	53.3	430.74	498.6	560.3	789.42	491.75	3350	187.09	454.23	518.21	578.44	806.62	515.11	3350
ITA	-8.97	411.36	483.43	547	873.68	475.66	11639	68.09	400.47	466.07	530.24	817.56	465.66	11639
LIE	250.65	467.06	529.87	588.42	725.95	525.08	332	222.67	469.86	539.2	608.6	799.88	535.8	332
LUX	57.92	415.74	488.15	551.42	742.46	479.42	3923	180.11	430.24	495.23	557.17	793.86	493.21	3923
LVA	119.81	430.95	494.94	554.26	782.02	490.56	4627	140.22	423.52	484.06	543.53	811.51	483.37	4627
NLD	204.49	454.11	516.93	576.14	759.73	513.12	3992	213.24	470.95	539.62	608.32	802.41	537.82	3992
NOR	47.28	434.25	507.11	570.56	813.38	499.74	4064	179.75	432.87	495.26	559.99	814.57	495.19	4064
POL	106.47	436.43	500.46	562.63	799.43	496.61	4383	136.97	428.2	489.67	552.81	776.63	490.24	4383
PRT	126.46	417.62	485.57	543.81	753.32	477.57	4608	170.06	405.97	467.18	526.14	751.77	466.02	4608
SVK	54.77	407.86	472.68	535.3	751.61	469.16	7346	115.78	435.55	498.33	564.59	825.27	498.18	7346
SWE	34.53	453.14	520.57	581.58	803.7	514.27	4624	117.85	446.08	509.6	575.57	799.64	509.05	4624

Note. The table presents the descriptive statistics on the reading and math test scores in 2003.

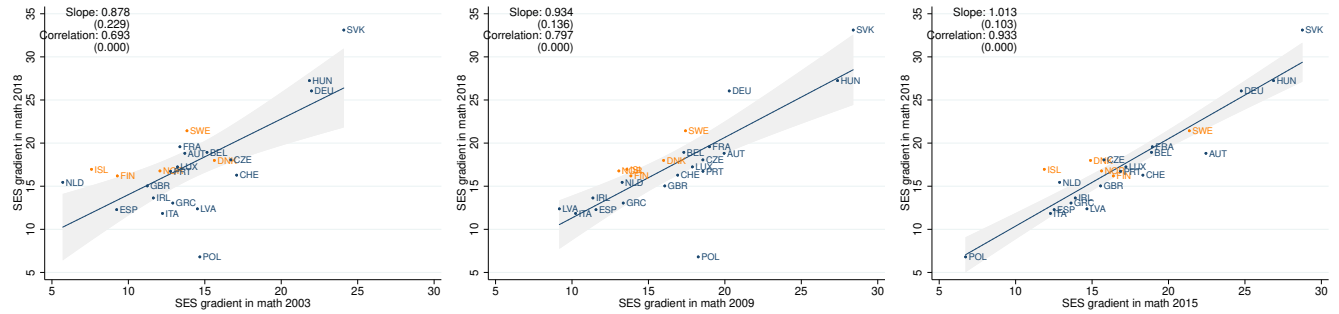
Table A6: Test score statistics, 2018

Country Name	Reading							Mathematics						
	Min	P25	P50	P75	Max	Mean	N	Min	P25	P50	P75	Max	Mean	N
AUT	163.36	412.73	488.16	557.72	771.94	484.39	6802	166.55	433.41	503.24	566.41	785.17	498.94	6802
BEL	145.69	421.39	497.6	567.82	801.94	492.86	8475	186.05	439.83	514.23	578.98	801.67	508.07	8475
CHE	70.28	412.76	488.08	557.74	796.57	483.93	5822	172.3	447.98	518.06	582.1	822.22	515.31	5822
CZE	164.68	422.06	491.61	559.9	804.6	490.22	7019	172.98	434.99	501.02	564.29	819.13	499.47	7019
DEU	164.65	424.21	503.81	575.87	836.03	498.28	5451	179.76	433.05	504.05	569.71	803.27	500.04	5451
DNK	130.35	438.56	504.41	566.12	791.55	501.13	7657	191.78	453.69	511.51	567.45	771.37	509.4	7657
ESP	147.93	413.24	479.21	542.51	799.77	476.54	35943	127.47	420.99	484.37	544.24	808.4	481.39	35943
FIN	145.86	455.09	526.9	591.18	823.9	520.08	5649	178.6	451.33	510.43	564.74	776.51	507.3	5649
FRA	139.57	422.89	496.94	566.62	807.17	492.61	6308	151.59	432.93	501.72	562.21	775.73	495.41	6308
GBR	160.47	435.36	506.3	574.83	846.54	503.93	13818	122.58	438.72	503.78	566.71	813.21	501.77	13818
GRC	157.19	390.2	459.73	526.46	771.41	457.41	6403	127.59	390.6	453.55	513.26	758.55	451.37	6403
HUN	180.88	406.69	479.06	547.24	776.83	475.99	5132	148.92	417.89	484.03	546.03	771.87	481.08	5132
IRL	207.82	456	519.95	582.64	797.02	518.08	5577	219.19	447.49	502.06	553.79	743.68	499.63	5577
ISL	163.82	402.48	477.44	549.44	795.8	473.97	3296	194.88	433.9	499.3	558.55	767.23	495.19	3296
ITA	142.5	413.14	481.34	544.63	785.98	476.28	11785	154.72	423.35	490.24	551.97	791.6	486.59	11785
LUX	153.3	392.36	471.54	547.95	804.29	469.99	5230	162.42	412.86	485.37	554.8	790.08	483.42	5230
LVA	170.92	415.21	480.49	542.35	775.1	478.7	5303	211.76	441.38	496.6	550.99	762.97	496.13	5303
NLD	129.98	410.44	486.41	562.18	794.29	484.78	4765	140.48	452.97	524.34	587.91	792.79	519.23	4765
NOR	153.17	430.32	506.15	575.84	810.56	499.45	5813	163.78	440.56	503.79	565.17	789.19	500.96	5813
POL	154.32	445.69	514.54	581.36	837.35	511.86	5625	192.84	454.78	516.78	577.96	810.78	515.65	5625
PRT	166.58	425.11	497.18	561.93	785.53	491.8	5932	158.4	426.04	497.29	561.71	799.26	492.49	5932
SVK	140.24	387.5	457.5	529.38	780.24	457.98	5965	119.08	419.55	491.72	556.45	801.07	486.16	5965
SWE	128	434.17	511.52	583.49	815.49	505.79	5504	193.5	440.51	504.89	566.84	788.26	502.39	5504

Note. The table presents the descriptive statistics on the reading and math test scores in 2018.

A.5 Trends in intergenerational transmission of learning by country

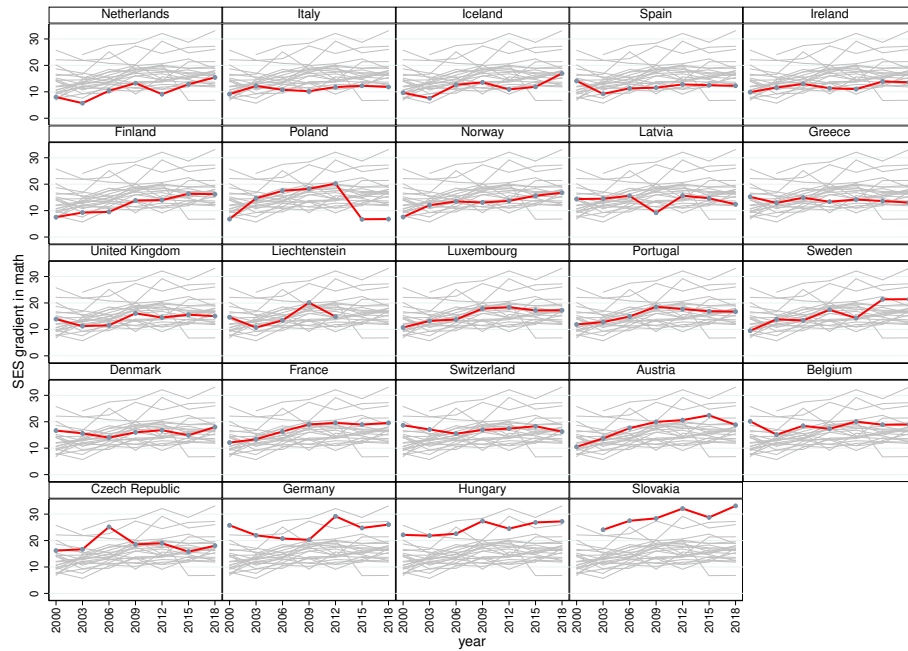
Figure A2: Correlation between SES gradients in math and its lags



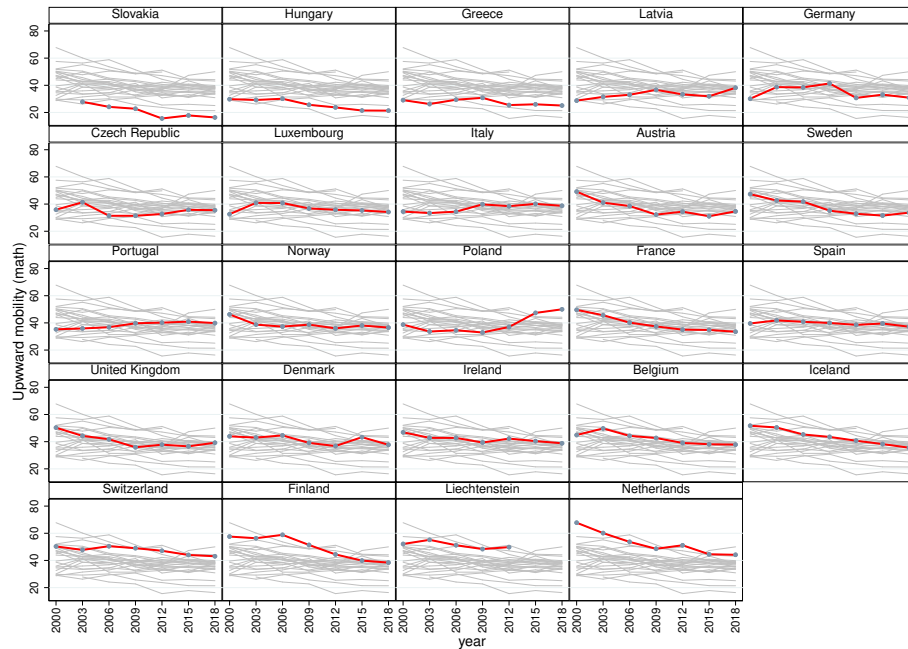
Note. The figures present the correlation between SES gradients in learning (math) in 2018 and its lags. The estimates are based on equation (1), regression of the rank math score on dummy equal to 1 if mother has at least completed upper secondary. The rank is computed at the European level. Higher values of SES gradients in learning correspond to lower mobility. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

Figure A3: Trends in the intergenerational transmission of learning (math) by country.

SES gradient in learning

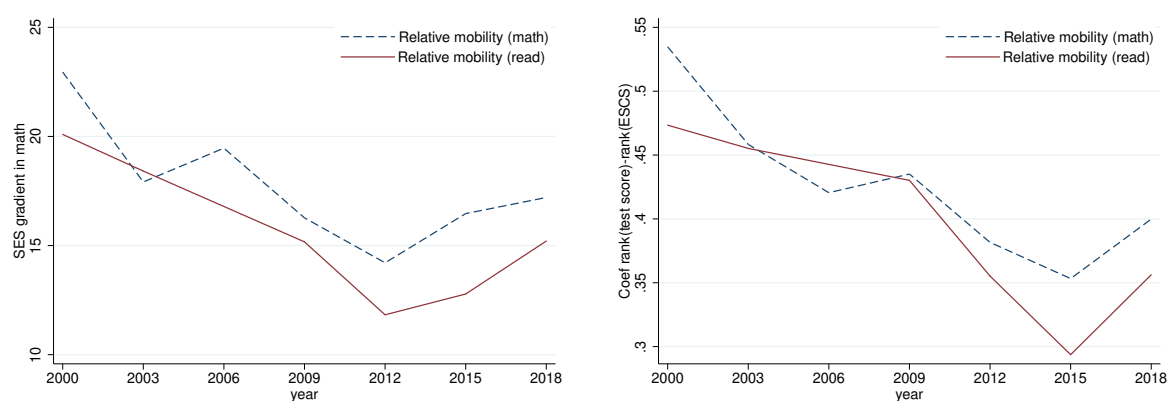


Upward Mobility



Note. The figures present the trends in the SES gradient and upward mobility for each country (red line) against other countries (gray lines). The estimates are based on equation (1), regression of the rank math score on dummy equal to 1 if mother has at least completed upper secondary. The rank is computed at the European level. Higher values of the SES gradient in learning correspond to lower mobility. Countries are ordered from the most to least mobile for SES gradient in learning, based on the country-average of SES gradients in learning over the PISA waves. Countries are ordered from the least to the most mobile for upward mobility, based on the country-average of upward mobility over the PISA waves.

Figure A4: Intergenerational transmission of learning in the United States.



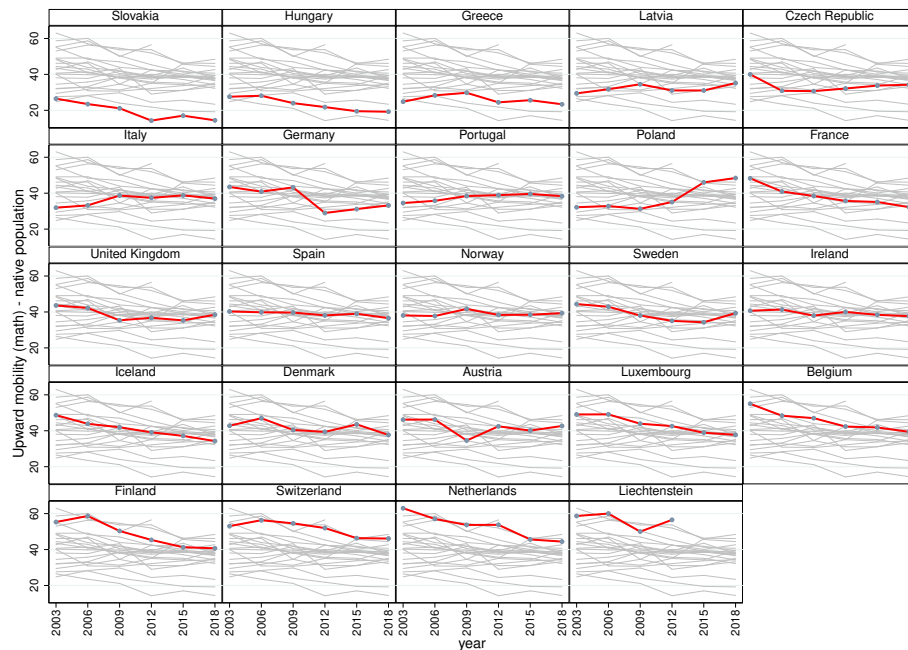
Note. The figures present the measures of intergenerational transmission of learning in the USA over time. The left panel presents the mobility measure computed by regressing the rank of the test score on a dummy equal to 1 if the mother has at least upper secondary (equation (1)). The right panel presents the mobility measure computed by regressing the rank of the test score on the rank of the ESCS index (equation (4)).

Figure A5: Trends in the intergenerational transmission of learning (math) by country (restricting sample to native population).

SES gradient in learning



Upward Mobility



Note. The figures present the trends in the SES gradient and upward mobility for each country (red line) against other countries (gray lines) restricting the sample to the native population. The estimates are based on equation (1) for the native population, regression of the rank math score on dummy equal to 1 if mother has at least completed upper secondary. The rank is computed at the European level for the native population (i.e., excluding the first- and second-generation immigrants). Immigrant status was not collected in the 2000 PISA wave. Countries are ordered from the most to least mobile for SES gradient in learning, based on the country-average of SES gradients in learning over the PISA waves. Countries are ordered from the least to the most mobile for upward mobility, based on the country-average of upward mobility over the PISA waves.

A.6 Do changes in composition drive changes in the intergenerational transmission of learning?

Table A7: intergenerational transmission of learning and migrant students

	SES gradient in learning		Upward mobility	
% migrant students	0.014 (0.074)	0.091 (0.110)	0.148 (0.144)	-0.126 (0.193)
Observations	142	142	142	142
R^2	0.001	0.819	0.036	0.829
Country FE	No	Yes	No	Yes
Cohort FE	No	Yes	No	Yes

Note. The table presents the regression of the SES gradient and upward mobility (math) on the proportion of first- and second-generation migrant students from the PISA database (2003-2018). The estimates of upward and SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. The estimates are presented with and without country and cohort fixed effects. Clustered standard errors at the country level (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$).

Table A8: intergenerational transmission of learning and mothers' education

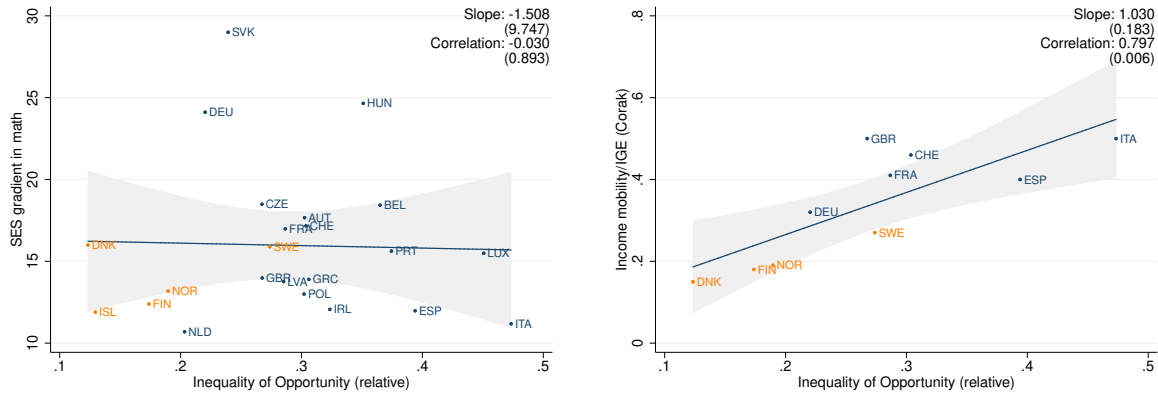
	SES gradient in learning		Upward mobility	
% mothers with less than upper secondary education	-0.107 (0.070)	-0.027 (0.105)	0.127 (0.084)	0.119 (0.172)
Observations	142	142	142	142
R^2	0.076	0.817	0.044	0.830
Country FE	No	Yes	No	Yes
Cohort FE	No	Yes	No	Yes

Note. The table presents the regression of the SES gradient and upward mobility (math) on the proportion of mothers with less than upper secondary education from the PISA database (2003-2018). The estimates of upward mobility and SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. The estimates are presented with and without country and cohort fixed effects. Clustered standard errors at the country level (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$).

B Correlates: intergenerational transmission of learning

B.1 Intergenerational transmission of learning and inequality of opportunity (relative)

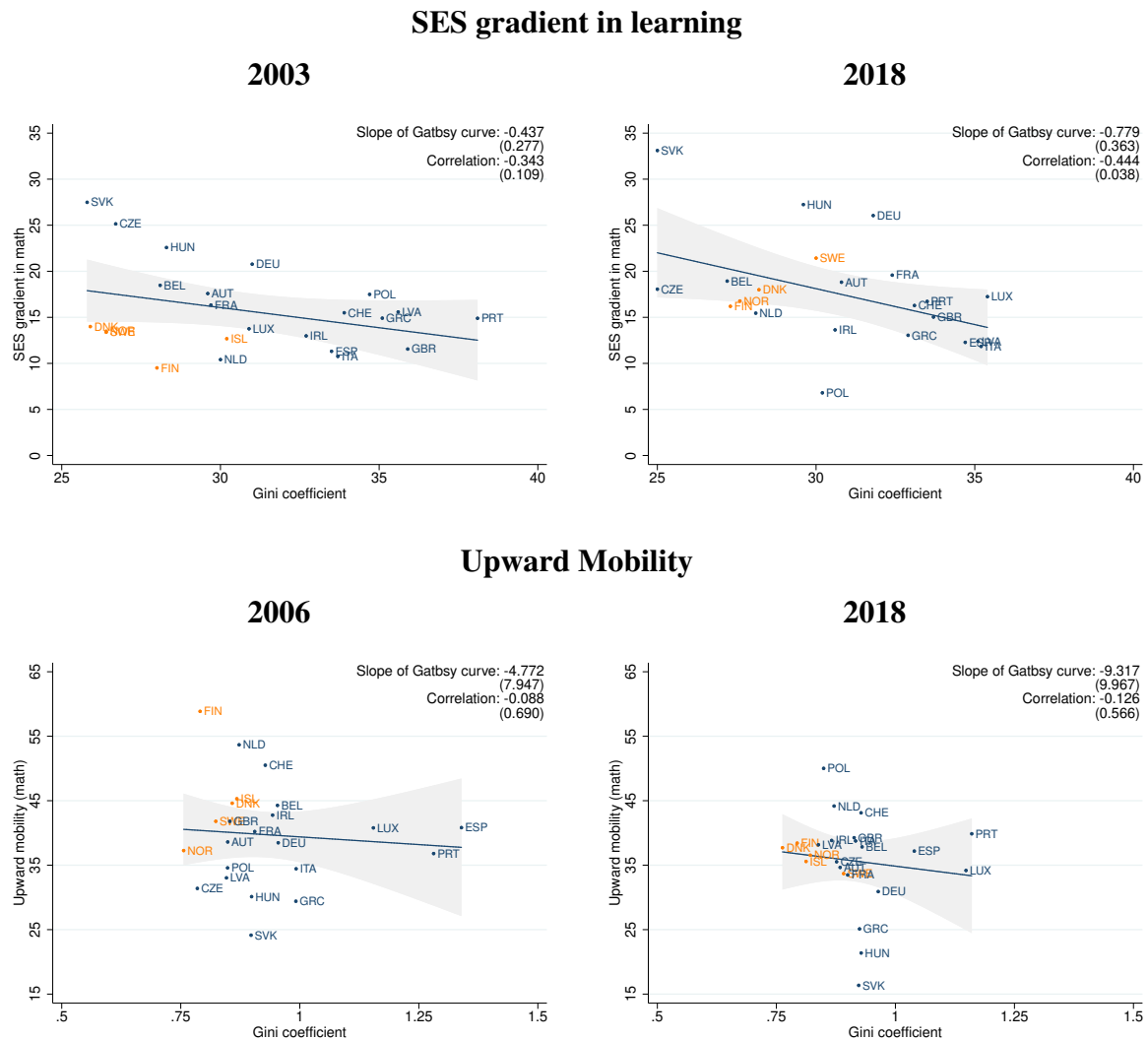
Figure B1: Correlation between SES gradients in learning, inequality of opportunity, IGE (Corak, 2013)



Note. The left figure presents the correlation between SES gradients in learning (math - averaged over PISA years) and inequality of opportunity (relative) in 2010 from <https://www.equalchances.org/>. The right figure presents the correlation between income mobility measure from Corak (2013) and inequality of opportunity (relative) from <https://www.equalchances.org/>. Corak (2013)'s estimates of income mobility are measured as the elasticity between paternal earnings and a son's adult earnings, using data on a cohort of children born during the early to mid 1960s and measuring their adult outcomes in the mid to late 1990s. These IGE estimates are derived from published studies, adjusted for methodological comparability in Corak (2013). Equality of opportunity captures to what extent individuals - exerting the same degree of effort (or responsibility) - attain the same level of advantage (or well-being), regardless of any predetermined circumstances beyond their control (Roemer, 1998). The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

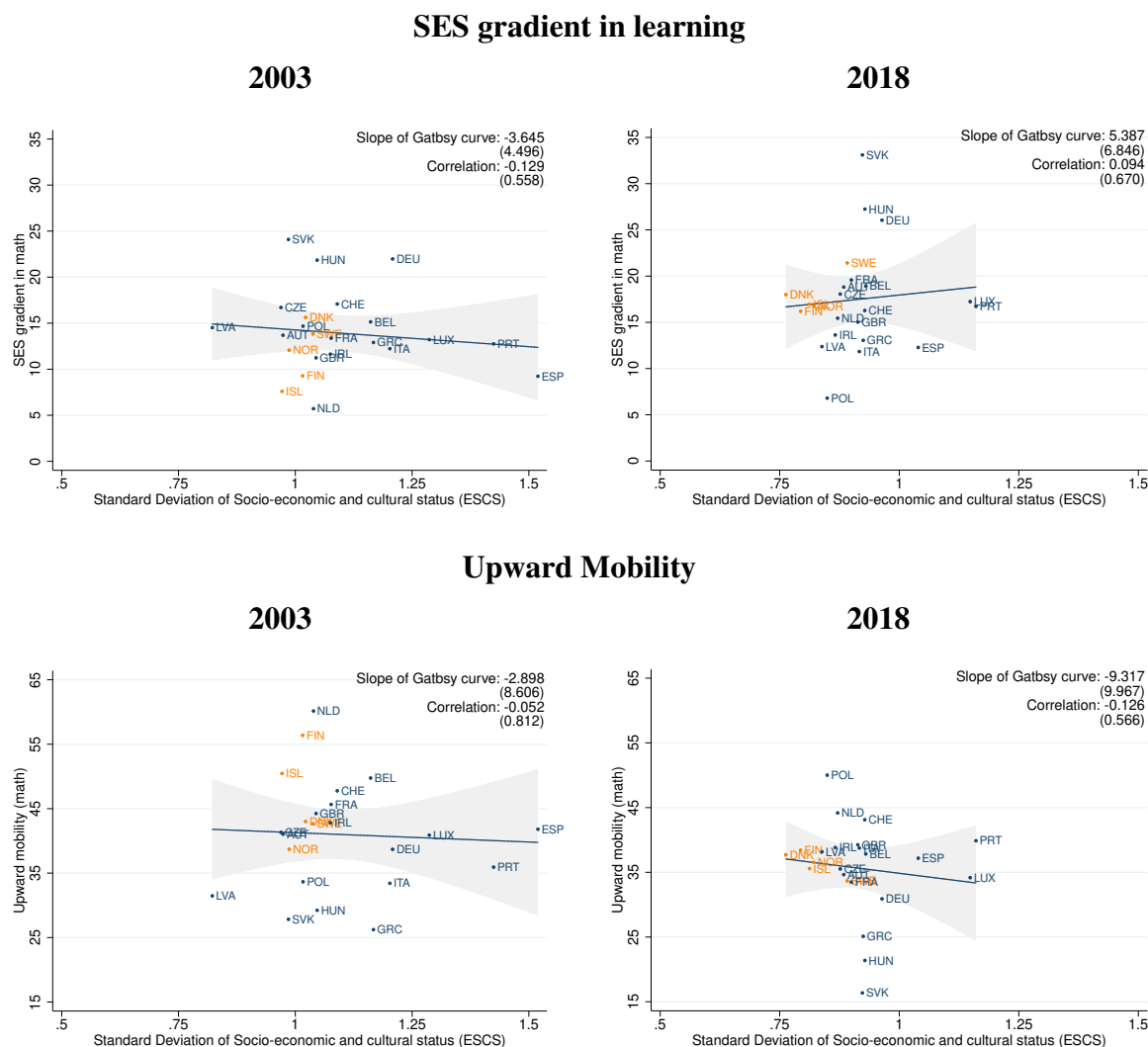
B.2 The Great Gatsby curve

Figure B2: The Great Gatsby curves: intergenerational transmission of learning and Gini coefficient (World Bank).



Note. The figures present the Gatsby curves by year. The estimates of SES gradients in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. The inequality measure is the Gini Coefficient from the World Bank database in the year of the PISA test. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

Figure B3: The Great Gatsby curves: intergenerational transmission of learning and standard deviation of ESCS index.



Note. The figures present the Gatsby curves by year. The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. The inequality measure is the standard deviation of ESCS index. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

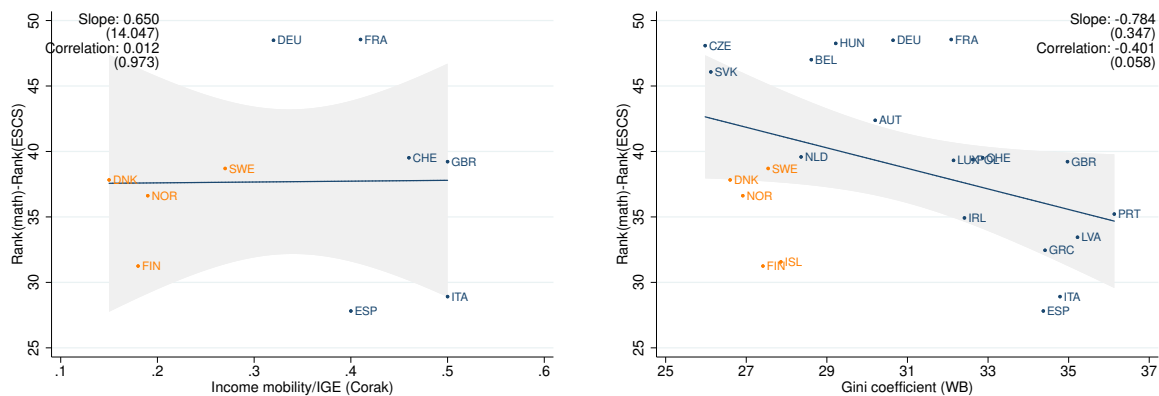
Table B1: Robustness to other definitions: intergenerational transmission of learning and intra/intergenerational inequality

Definition	SES gradient in learning					
	$\beta_{tc}^{math-father}$	β_{tc}^{read}	β_{tc}^{math-c}	$\beta_{tc}^{math-uni}$	$\beta_{tc}^{math-med}$	ρ_{tc}^{math}
Gini coefficient (WB)	-0.502 (0.316)	-0.367 (0.242)	-0.458 (0.280)	-0.032 (0.201)	-0.102 (0.207)	-0.784** (0.347)
Observations	23	23	23	23	23	23
R^2	0.105	0.083	0.115	0.001	0.010	0.161

Definition	SES gradient in learning					
	$\beta_{tc}^{math-father}$	β_{tc}^{read}	β_{tc}^{math-c}	$\beta_{tc}^{math-uni}$	$\beta_{tc}^{math-med}$	ρ_{tc}^{math}
Income mobility/IGE (Corak)	2.609 (7.610)	-0.975 (6.550)	-3.730 (6.063)	-6.560 (4.542)	2.577 (6.121)	0.650 (14.047)
Observations	10	10	10	10	10	10
R^2	0.007	0.001	0.022	0.147	0.015	0.000

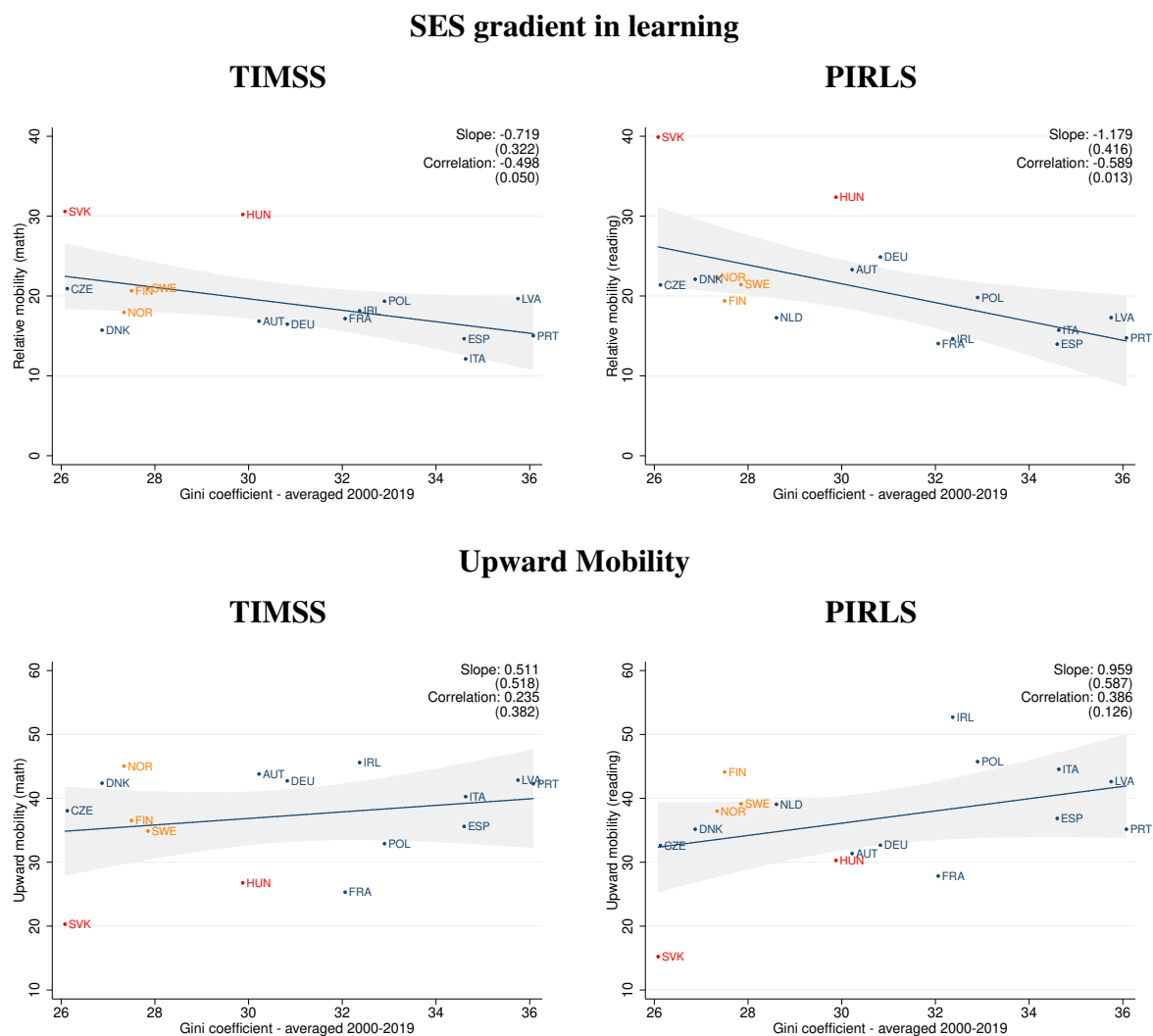
Note. The table presents the robustness to other definitions of intergenerational transmission of learning for the relationship between intergenerational transmission of learning and intra/intergenerational inequality. The other measures are defined in Appendix A.3 and reported in the note of the Table. $\beta_{tc}^{math-father}$ is the measure of the SES gradient in learning in math from equation (1) when using the father's education dummy instead of the mother's, β_{tc}^{read} is the main measure of the SES gradient in reading from equation (1), β_{tc}^{math-c} is the SES gradient in math from equation (1) when rank is computed within the country, $\beta_{tc}^{math-uni}$ is the SES gradient in math from equation (3), $\beta_{tc}^{math-med}$ is the SES gradient in math from equation (5), ρ_{tc}^{math} is the SES gradient from equation (4) using the rank of the parents' ESCS. Robust standard errors (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure B4: Correlation between rank(math)-rank(ESCS), IGE (Corak, 2013), and Gini coefficient



Note. The Left Figure presents the correlation between the coefficient from the rank(math)-rank(ESCS) regression (averaged over PISA years) and income mobility measure from Corak (2013). Corak (2013)'s estimates of income mobility are measured as the elasticity between paternal earnings and a son's adult earnings, using data on a cohort of children born during the early to mid 1960s and measuring their adult outcomes in the mid to late 1990s. These IGE estimates are derived from published studies, adjusted for methodological comparability in Corak (2013). The right figure presents the correlation between the coefficient from the rank(math)-rank(ESCS) regression and Gini coefficient from the World Bank (averaged over all PISA years). The estimates of the coefficient from the rank(math)-rank(ESCS) regression are based on a regression of the rank of the math score on the rank of ESCS. The rank is computed at the European level. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

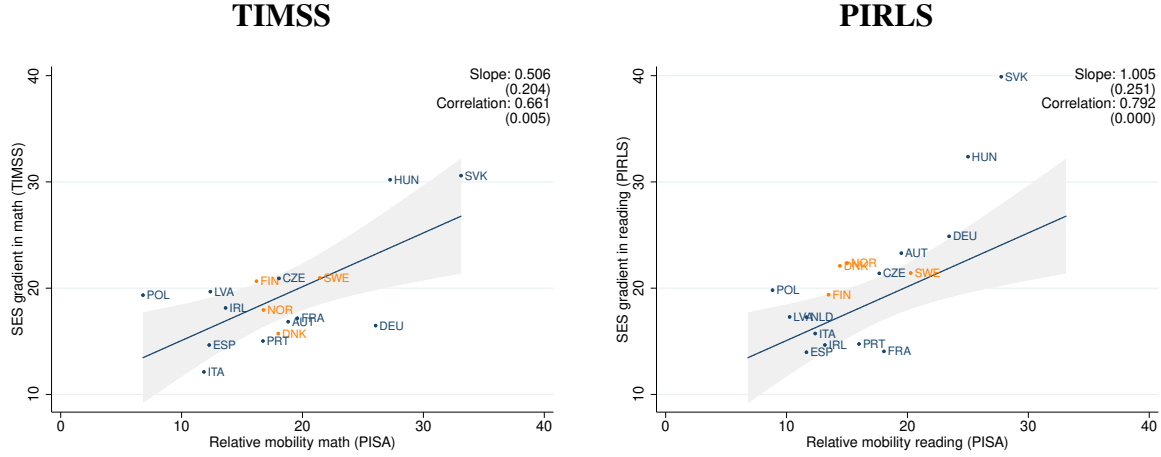
Figure B5: The Great Gatsby curves: intergenerational transmission of learning from TIMSS & PIRLS and Gini coefficient (World Bank).



Note. The figures present the Gatsby curves using the TIMSS and PIRLS data. The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score (4-grade 2019 TIMSS)/ reading score (4-grade 2016 PIRLS) on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. The inequality measure is the Gini Coefficient from the World Bank database in the year of the PISA test. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

B.3 Correlation between intergenerational transmission of learning in PISA and TIMSS/PIRLS

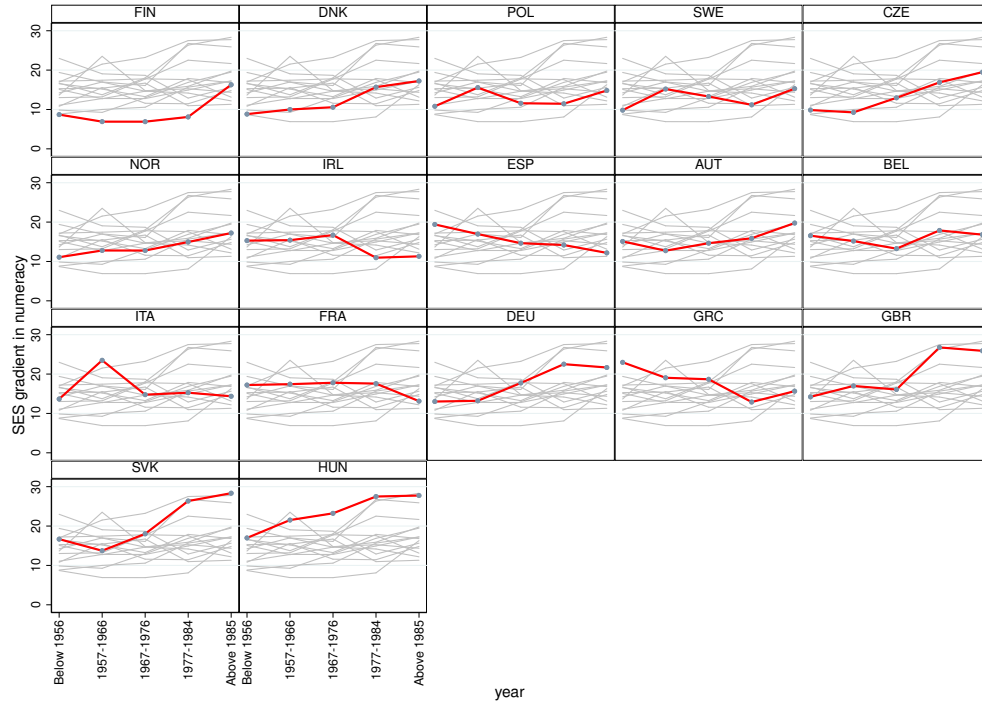
Figure B6: Correlation between intergenerational transmission of learning in PISA and TIMSS/PIRLS



Note. The figure presents the correlation between intergenerational transmission of learning in PISA and TIMSS. The mobility measure is the β_{tc}^k of regressing the rank of math skill on a dummy equal to 1 if the mother has at least upper secondary ((1)). The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score (4-grade 2019 TIMSS)/ reading score (4-grade 2016 PIRLS) on a dummy equal to 1 if the mother has at least upper secondary. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

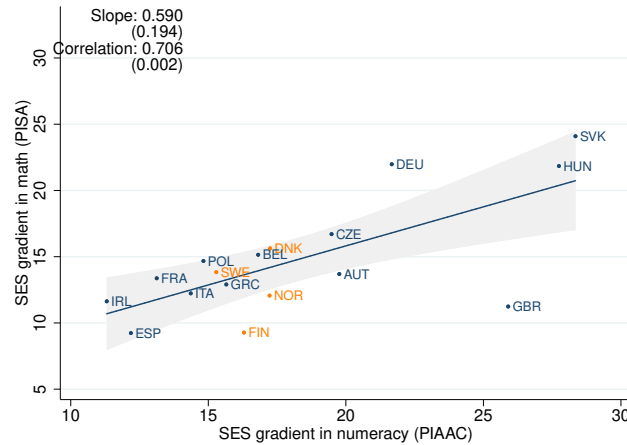
B.4 Evidence from PIAAC data

Figure B7: Trends in SES gradients in numeracy skill by cohort (PIAAC data).



Note. The figure presents the trends in SES gradients in numeracy skill by cohort. The mobility measure is the β_{tc}^k of regressing the rank of numeracy skill on a dummy equal to 1 if the mother has at least upper secondary. The rank is at European and cohort level.

Figure B8: Correlation between intergenerational transmission of learning in PISA and PIAAC



Note. The figure presents the scatter plots between SES gradients in math (PISA) and SES gradients in numeracy (PIAAC) for the 2003 PISA cohort (i.e., the cohort overlapping with PIAAC one - born 1987 or later). Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the p-value in parenthesis.

B.5 SES gradients in parental investment

B.5.1 Measuring parental investment

To construct a measure of parental investment in home educational resources that is comparable across countries and over time, we use a factor model specified in equation (6).

Assuming each latent item, $m_{ict}^{*,j}$ for question j , is additively separable in the logarithm of the latent factor, we have:

$$m_{ict}^{*,j} = \alpha_{ct}^j + \lambda_t^{j\top} \ln I_{ict} + \varepsilon_{ict}^j \quad (6)$$

Depending on the nature of $m_{ict}^{*,j}$, we need to specify different models:

- (i) Continuous variables: $m_{ict-1} = m_{ict-1}^*$;
- (ii) Binary variables $m_{ict-1} \in \{0, 1\}$: $Prob\{m_{ict-1} = 1\} = Pr\{m_{ict-1}^* \geq 0\}$;
- (iii) Categorical variables $m_{ict-1} \in \{1, 2, \dots, L\}$: $Prob\{m_{ict-1} = l\} = Pr\{\tau_{l-1} \leq m_{ict-1}^* \leq \tau_l\}$, where $\tau_0 = -\infty$;

We use the threshold model defined in (iii) as items are categorical. Given that factors are unobserved and lack a scale, ensuring comparability requires some identification. To achieve this, we fix the scale and location based on the same item $j = 1$ across countries and over time. Namely, we set $\lambda_{ct}^1 = 1$ and $\tau_{c1,t}^1 = 0$ and $\alpha_{ct}^j = 0, \forall j$ respectively. We use the following item "Possessions poetry" which has been asked across all the waves and countries. We use the weights provided by the OECD throughout the analysis to estimate the factor model.

The questions used for the parental investment index have been collected from 2003 and are presented below:³

- Possessions desk
- Possessions study place
- Possessions computer
- Possessions software
- Possessions textbooks
- Possessions <technical reference books>

³More details on the questions can be found in the technical reports. Technical report 2003: page 283 <https://www.oecd.org/education/school/programme-for-international-student-assessment-pisa/35188570.pdf> Technical report 2006: Page 316 <https://www.oecd.org/pisa/data/42025182.pdf> Technical report 2009: page 288 <https://www.oecd.org/pisa/pisaproducts/50036771.pdf> Technical report 2012: page 316 <https://www.oecd.org/pisa/pisaproducts/PISA-2012-technical-report-final.pdf> Technical report 2015: page 304 https://www.oecd.org/pisa/data/2015-technical-report/PISA2015_TechRep_Final.pdf

- Possessions dictionary
- Possessions literature
- Possessions poetry
- Possessions art
- How many books at home

For each student i , country c and cohort t , we estimate:

$$R(I_{itc}) = \alpha_{tc}^I + \beta_{tc}^I HS_{itc}^P + e_{itc} \quad (7)$$

where $R(I_{itc})$ is the parental investment percentile rank (rank computed at the European level) for student i , in cohort t , living in country c . HS_{itc}^P is an indicator variable taking the value 1 if i 's mother has completed at least upper secondary education, and 0 otherwise (Appendix Table A2).

β_{tc}^I measures the SES gradient in parental investment; i.e., the association between mother's education (high school degree) and her child's rank in the distribution of parental investment, for cohort t and country c .

α_{tc}^I measures the average level of investment in children whose mothers do not have upper secondary education.

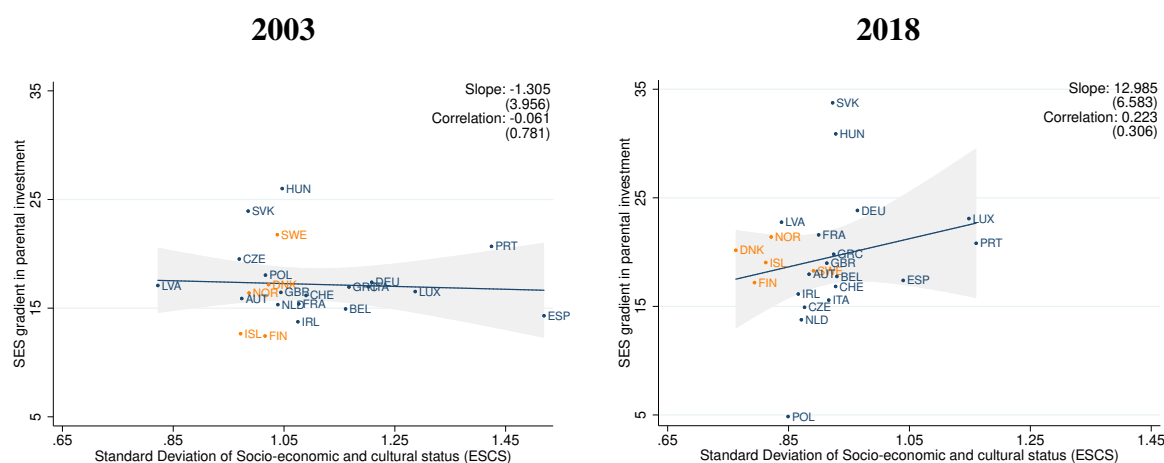
B.5.2 Correlates of the socio-economic gradients in parental investment

Appendix Figure B9 shows a lack of correlation between inequality in ESCS and SES gradients in parental investments, hinting that inequality in the ESCS and SES gradients in parental investment are measuring two different factors.

Appendix Figure B10 explores to what extent attitudes towards education correlate with SES gradients in parental investment, aiming to investigate the underlying mechanism. We use the data from the European Value Study (EVS) in 2008 and 2017 and the International Social Survey Programme (ISSP) on social inequality in 2009. For the EVS, the question is "How much confidence you have in the education system, is it a great deal, quite a lot, not very much or none at all? (Answers: 1. a great deal, 2. quite a lot, 3. not very much, 4. none at all). We reverse code the variable and recode "cannot choose" as missing (few people answer cannot choose). We then collapse the data to obtain the mean by country, using the weight provided by EVS.

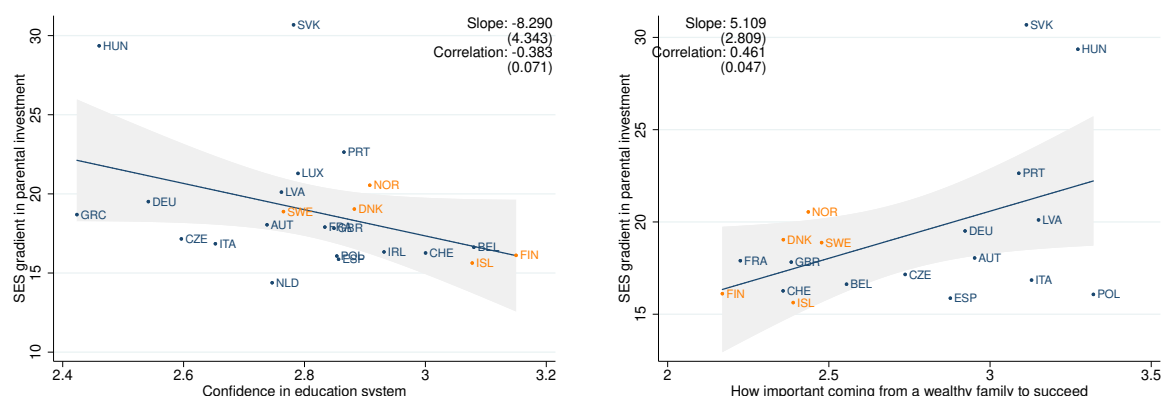
For the ISSP, the question is "How important you think is coming from a wealthy family for getting ahead in life?" (Answers: 1. Essential, 2. Very Important, 3 Fairly Important, 4. Not very important, 5. Not important at all, 8. cannot choose). We reverse code the variable and recode "cannot choose" as missing (few people answer cannot choose). We then collapse the data to obtain the mean by country, using the weight provided by ISSP.

Figure B9: Correlation between SES gradients in parental investment and inequality in ESCS.



Note. The figures present the correlation between the SES gradients in parental investment and inequality in ESCS. The estimates of the socio-economics gradients in parental investment are based on regression of the rank of the parental investment on a dummy equal to 1 if the mother has at least upper secondary (equation (7)). The rank is computed at the European level. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the p-value in parenthesis.

Figure B10: Correlation between the SES gradients in parental investment and attitudes towards education.



Note. The figures present the correlation between the socio-economics gradients in parental investment (averaged over 2003-2018 PISA waves) and attitudes towards education. The estimates of the socio-economics gradients in parental investment are based on regression of the rank of the parental investment on a dummy equal to 1 if the mother has at least upper secondary (equation (7)). The rank is computed at the European level. For the EVS, the question is "How much confidence you have in the education system? (answers: 1. none at all, 2. not very much, 3. quite a lot, 4. a great deal). For the ISSP, the question is "How important you think is coming from a wealthy family for getting ahead in life?" (answers: 1. Not important at all, 2. Not very important, 3 Fairly Important, 4. Very Important, 5. Essential). Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the p-value in parenthesis.

B.6 Robustness: intergenerational transmission of learning and SES gradients in parental investment

Table B2: Cross-section regressions: SES gradient in test scores - Standardized coeffs

	SES gradient in math							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log GDP per capita	-0.199 (0.167)							-0.253 (0.240)
Returns to secondary school		0.450* (0.217)						0.007 (0.154)
Difference between top and lowest tax rates on personal income			-0.187 (0.265)					0.128 (0.166)
Government expenditure on education (% of GDP)				-0.146 (0.167)				0.262 (0.172)
School tracking age					-0.490*** (0.158)			-0.525** (0.224)
School segregation						0.606*** (0.171)		0.029 (0.266)
SES gradient in parental investment							0.755*** (0.115)	0.518** (0.192)
Observations	24	22	23	23	23	23	24	20
R ²	0.040	0.193	0.034	0.020	0.369	0.352	0.570	0.739

Note. The table presents the cross-sectional estimates of the SES gradient in learning on economic variables with the standardized coefficients. Variables are standardized to have mean 0 and standard deviation 1. The estimates of upward mobility and SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Robust standard errors (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B3: Robustness to outliers in the cross section: intergenerational transmission of learning and SES gradients in parental investment

	SES gradient in learning (math)					
Excluding	AUT	BEL	CHE	CZE	DEU	DNK
SES gradient in parental investment	0.845*** (0.129)	0.866*** (0.124)	0.864*** (0.125)	0.858*** (0.127)	0.825*** (0.140)	0.841*** (0.128)
Observations	23	23	23	23	23	23
R ²	0.579	0.605	0.593	0.598	0.643	0.570
Excluding	ESP	FIN	FRA	GBR	GRC	HUN
SES gradient in parental investment	0.828*** (0.132)	0.831*** (0.132)	0.845*** (0.129)	0.838*** (0.128)	0.840*** (0.127)	0.849*** (0.187)
Observations	23	23	23	23	23	23
R ²	0.560	0.561	0.576	0.569	0.575	0.483
Excluding	IRL	ISL	ITA	LIE	LUX	LVA
SES gradient in parental investment	0.828*** (0.131)	0.828*** (0.133)	0.824*** (0.130)	0.879*** (0.121)	0.859*** (0.123)	0.853*** (0.123)
Observations	23	23	23	23	23	23
R ²	0.563	0.558	0.570	0.586	0.584	0.590
Excluding	NLD	NOR	POL	PRT	SVK	SWE
SES gradient in parental investment	0.821*** (0.137)	0.862*** (0.119)	0.836*** (0.131)	0.881*** (0.114)	0.678*** (0.150)	0.841*** (0.128)
Observations	23	23	23	23	23	23
R ²	0.548	0.604	0.562	0.599	0.362	0.570

Note. The table presents the robustness to outliers for intergenerational transmission of learning and SES gradients in parental investment. The relationship between intergenerational transmission of learning and SES gradients in parental investment is re-estimated by excluding each country one by one. The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Robust standard errors (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B4: Robustness to compositional changes: cross section

SES gradient in learning					
SES gradient in parental investment	0.841*** (0.128)	0.803*** (0.108)	0.870*** (0.146)	0.836*** (0.127)	0.661** (0.240)
% migrant students			0.057 (0.083)	0.075 (0.083)	-0.029 (0.104)
% mothers with less than upper secondary education		-0.070** (0.032)		-0.079** (0.029)	-0.129*** (0.042)
Observations	24	24	24	24	21
R^2	0.570	0.608	0.585	0.633	0.758
Other controls	No	No	No	No	Yes
Upward mobility					
SES gradient in parental investment	-1.483*** (0.174)	-1.461*** (0.182)	-1.427*** (0.185)	-1.415*** (0.197)	-1.057*** (0.242)
% migrant students			0.109 (0.073)	0.103 (0.077)	-0.087 (0.152)
% mothers with less than upper secondary education		0.043 (0.069)		0.029 (0.076)	0.069 (0.091)
Observations	24	24	24	24	21
R^2	0.646	0.651	0.666	0.668	0.781
Other controls	No	No	No	No	Yes

Note. The table shows that the association in the cross section between the SES gradient in learning outcomes and the SES gradient in parental investment remains robust, when controlling for compositional factors, such as the percentage of first/second-generation migrant students and the percentage of mothers with less than a high school education. Other controls are: *socio-economic variables* (returns to secondary school from the OECD, and country GDP per capita from the World Bank database); *institutional variables* (the progressivity of the tax system, proxied by the difference between the top and lowest tax rates on personal income, and spending in public education from the World Bank database) and *education system variables* (age at which academic tracking starts from Eurydice and school level segregation, which we measure from the PISA data as the R-squared from regressing the ESCS index on school fixed effects). Robust standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$).

Table B5: Robustness to other definitions in the cross section: intergenerational transmission of learning and SES gradients in parental investment

SES gradient in learning						
Panel A Definition	$\beta_{tc}^{math-father}$	β_{tc}^{read}	β_{tc}^{math-c}	$\beta_{tc}^{math-uni}$	$\beta_{tc}^{math-med}$	ρ_{tc}^{math}
SES gradient in parental investment	0.935*** (0.096)	0.714*** (0.105)	1.022*** (0.153)	1.132*** (0.211)	0.445*** (0.144)	35.650** (15.735)
Observations	24	24	24	24	24	23
R^2	0.687	0.492	0.649	0.631	0.275	0.198
Other controls	No	No	No	No	No	No
SES gradient in learning						
Panel B						
SES gradient in parental investment	0.378*** (0.124)	0.457*** (0.137)	0.707*** (0.211)	0.637** (0.227)	-0.022 (0.125)	11.661 (9.741)
Observations	74	74	74	74	74	74
R^2	0.925	0.943	0.927	0.925	0.835	0.907
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Upward mobility						
Panel C Definition	$\alpha_{tc}^{math-father}$	α_{tc}^{read}	α_{tc}^{math-c}	$\alpha_{tc}^{math-uni}$	$\alpha_{tc}^{math-med}$	μ_{tc}^{math}
SES gradient in parental investment	-1.381*** (0.124)	-1.375*** (0.231)	-1.000*** (0.256)	-0.924** (0.377)	-0.941*** (0.167)	-15.093* (8.293)
Observations	24	24	24	24	24	23
R^2	0.662	0.626	0.517	0.164	0.326	0.074
Other controls	No	No	No	No	No	No
Upward mobility						
Panel D						
SES gradient in parental investment	-1.150*** (0.370)	-0.607* (0.327)	-0.687* (0.328)	-0.085 (0.488)	-0.944** (0.380)	-45.847** (15.181)
Observations	20	20	20	20	20	20
R^2	0.717	0.627	0.611	0.573	0.753	0.544
Other controls	Yes	Yes	Yes	Yes	Yes	Yes

Note. The table presents the robustness in the cross section to other definitions of intergenerational transmission of learning for the relationship between intergenerational transmission of learning and SES gradients in parental investment. The other measures are defined in Appendix A.3 and reported in the note of the table. The socio-economic status variable is defined in the same way for the SES gradient in test scores and SES gradient in parental investment. $\beta_{tc}^{math-father}$ is the measure of the SES gradient in math from equation (1) when using the father's education dummy instead of the mother's, β_{tc}^{read} is the main measure of the SES gradient in reading from equation (1), β_{tc}^{math-c} is the SES gradient in math from equation (1) when rank is computed within the country, $\beta_{tc}^{math-uni}$ is the SES gradient in math from equation (3), $\beta_{tc}^{math-med}$ is the SES gradient in math from equation (5), ρ_{tc}^{math} is the SES gradient from equation (4) using the rank of the parents' ESCS. α_{tc}^{math} is the main measure of upward mobility in math from equation (1), $\alpha_{tc}^{math-father}$ is the main measure of upward mobility in math from equation (1) when using the father's education dummy instead of the mother's, α_{tc}^{read} is the main measure of upward mobility in reading from equation (1), α_{tc}^{math-c} is upward mobility in math from equation (1) when rank is computed within the country, $\alpha_{tc}^{math-uni}$ is upward mobility in math from equation (3), $\alpha_{tc}^{math-med}$ is upward mobility in math from equation (5), μ_{tc}^{math} is upward mobility from equation (4) using the rank of the parents' ESCS. Robust standard errors (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B6: Panel regressions: SES gradient in test scores - Standardized coeffs

	SES gradient in math						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log GDP per capita	0.705 (0.441)						-0.101 (0.809)
Returns to secondary school		0.162* (0.081)					0.059 (0.125)
Difference between top and lowest tax rates on personal income			0.097 (0.223)				0.297 (0.382)
Government expenditure on education (% of GDP)				-0.086 (0.143)			-0.054 (0.145)
School segregation					0.118 (0.170)		0.009 (0.271)
SES gradient in parental investment						0.513*** (0.117)	0.592*** (0.195)
Observations	160	118	137	117	159	142	74
R^2	0.805	0.841	0.808	0.819	0.802	0.881	0.914
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note. The table presents the panel estimates of the SES gradient in learning on economic variables with year and country fixed effects with the standardized coefficients. Variables are standardized to have mean 0 and standard deviation 1. The estimates of upward mobility and SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Clustered standard errors at the country level (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$).

Table B7: Robustness to outliers in the panel regression: intergenerational transmission of learning and SES gradients in parental investment

	SES gradient in learning (math)					
Excluding	AUT	BEL	CHE	CZE	DEU	DNK
SES gradient in parental investment	0.557*** (0.131)	0.572*** (0.127)	0.560*** (0.133)	0.552*** (0.146)	0.555*** (0.141)	0.565*** (0.131)
Observations	136	136	136	136	136	136
R^2	0.882	0.882	0.883	0.892	0.873	0.883
Excluding	ESP	FIN	FRA	GBR	GRC	HUN
SES gradient in parental investment	0.568*** (0.129)	0.554*** (0.132)	0.575*** (0.127)	0.571*** (0.129)	0.570*** (0.129)	0.563*** (0.131)
Observations	136	136	136	136	136	136
R^2	0.877	0.880	0.883	0.881	0.884	0.863
Excluding	IRL	ISL	ITA	LIE	LUX	LVA
SES gradient in parental investment	0.564*** (0.131)	0.565*** (0.131)	0.560*** (0.132)	0.553*** (0.133)	0.569*** (0.131)	0.573*** (0.131)
Observations	136	136	136	138	136	136
R^2	0.880	0.882	0.879	0.884	0.882	0.894
Excluding	NLD	NOR	POL	PRT	SVK	SWE
SES gradient in parental investment	0.590*** (0.117)	0.581*** (0.122)	0.392*** (0.112)	0.572*** (0.129)	0.607*** (0.122)	0.593*** (0.113)
Observations	136	136	136	136	136	136
R^2	0.887	0.885	0.887	0.883	0.841	0.897

Note. The table presents the robustness in the panel to outliers for intergenerational transmission of learning and SES gradients in parental investment. The relationship between intergenerational transmission of learning and SES gradients in parental investment is re-estimated by excluding each country one by one. The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Clustered standard errors at the country level (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$).

Table B8: Robustness to compositional changes: panel regression

SES gradient in learning					
SES gradient in parental investment	0.564*** (0.129)	0.579*** (0.120)	0.561*** (0.127)	0.575*** (0.120)	0.682*** (0.189)
% migrant students			0.034 (0.090)	0.025 (0.083)	-0.168 (0.171)
% mothers with less than upper secondary education		-0.079 (0.067)		-0.078 (0.065)	-0.136 (0.153)
Observations	142	142	142	142	74
R ²	0.881	0.884	0.882	0.884	0.919
Other controls	No	No	No	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes
Upward mobility					
SES gradient in parental investment	-0.750*** (0.229)	-0.783*** (0.212)	-0.744*** (0.227)	-0.780*** (0.212)	-0.954*** (0.230)
% migrant students			-0.050 (0.161)	-0.026 (0.149)	0.464* (0.256)
% mothers with less than upper secondary education		0.189 (0.131)		0.188 (0.130)	0.432* (0.209)
Observations	142	142	142	142	74
R ²	0.874	0.880	0.874	0.880	0.935
Other controls	No	No	No	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes

Note. The table shows that the association in the panel between the SES gradient in learning outcomes and the SES gradient in parental investment remains robust, when controlling for compositional factors, such as the percentage of first/second-generation migrant students and the percentage of mothers with less than a high school education. Other controls are: *socio-economic variables* (returns to secondary school from the OECD, and country GDP per capita from the World Bank database); *institutional variables* (the progressivity of the tax system, proxied by the difference between the top and lowest tax rates on personal income, and spending in public education from the World Bank database) and *education system variables* (school level segregation, which we measure from the PISA data as the R-squared from regressing the ESCS index on school fixed effects). Clustered standard errors at the country level (*** p<0.01, ** p<0.05, * p<0.1).

Table B9: Robustness to other definitions in the panel regression: intergenerational transmission of learning and SES gradients in parental investment

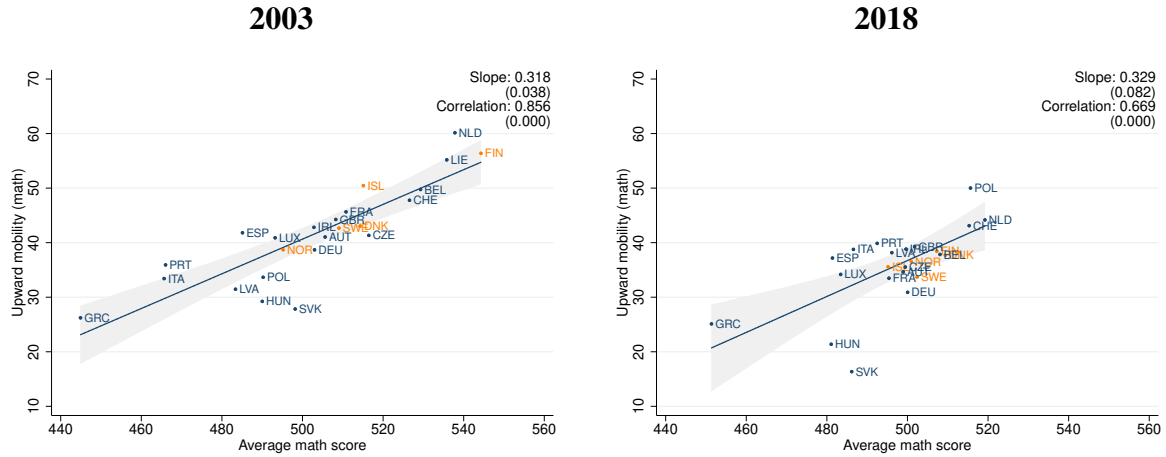
SES gradient in learning						
Panel A						
Definition	$\beta_{tc}^{math-father}$	β_{tc}^{read}	β_{tc}^{math-c}	$\beta_{tc}^{math-uni}$	$\beta_{tc}^{math-med}$	ρ_{tc}^{math}
SES gradient in parental investment	0.438*** (0.087)	0.502*** (0.118)	0.634*** (0.124)	0.687*** (0.111)	0.138** (0.065)	18.020 (11.711)
Observations	142	142	142	142	142	137
R^2	0.889	0.880	0.889	0.857	0.681	0.885
Other controls	No	No	No	No	No	No
Panel B						
SES gradient in parental investment	0.378*** (0.124)	0.457*** (0.137)	0.707*** (0.211)	0.637** (0.227)	-0.022 (0.125)	11.661 (9.741)
Observations	74	74	74	74	74	74
R^2	0.925	0.943	0.927	0.925	0.835	0.907
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Upward mobility						
Panel C						
Definition	$\alpha_{tc}^{math-father}$	α_{tc}^{read}	α_{tc}^{math-c}	$\alpha_{tc}^{math-uni}$	$\alpha_{tc}^{math-med}$	μ_{tc}^{math}
SES gradient in parental investment	-0.522*** (0.160)	-0.596*** (0.167)	-0.612*** (0.149)	-0.276 (0.190)	-0.285* (0.139)	-10.382 (10.635)
Observations	142	142	142	142	142	137
R^2	0.883	0.878	0.903	0.854	0.811	0.749
Other controls	No	No	No	No	No	No
Panel D						
SES gradient in parental investment	-0.549** (0.220)	-0.416* (0.236)	-0.709*** (0.229)	-0.126 (0.293)	-0.094 (0.217)	-6.114 (15.225)
Observations	74	74	74	74	74	74
R^2	0.932	0.931	0.935	0.881	0.892	0.859
Other controls	Yes	Yes	Yes	Yes	Yes	Yes

Note. The table presents the robustness in the panel to other definitions of intergenerational transmission of learning for the relationship between intergenerational transmission of learning and SES gradients in parental investment. The other measures are defined in Appendix A.3 and reported in the note of the table. The socio-economic status variable is defined in the same way for the SES gradient in test scores and SES gradient in parental investment. $\beta_{tc}^{math-father}$ is the measure of the SES gradient in math from equation (1) when using the father's education dummy instead of the mother's, β_{tc}^{read} is the main measure of the SES gradient in reading from equation (1), β_{tc}^{math-c} is the SES gradient in math from equation (1) when rank is computed within the country, $\beta_{tc}^{math-uni}$ is the SES gradient in math from equation (3), $\beta_{tc}^{math-med}$ is SES gradient in math from equation (5), ρ_{tc}^{math} is the SES gradient from equation (4) using the rank of the parents' ESCS. α_{tc}^{math} is the main measure of upward mobility in math from equation (1), $\alpha_{tc}^{math-father}$ is the main measure of upward mobility in math from equation (1) when using the father's education dummy instead of the mother's, α_{tc}^{read} is the main measure of upward mobility in reading from equation (1), α_{tc}^{math-c} is upward mobility in math from equation (1) when rank is computed within the country, $\alpha_{tc}^{math-uni}$ is upward mobility in math from equation (3), $\alpha_{tc}^{math-med}$ is upward mobility in math from equation (5), μ_{tc}^{math} is upward mobility from equation (4) using the rank of the parents' ESCS. Clustered standard errors at the country level (*** p<0.01, ** p<0.05, * p<0.1).

C Other measures of SES gradients in child outcomes

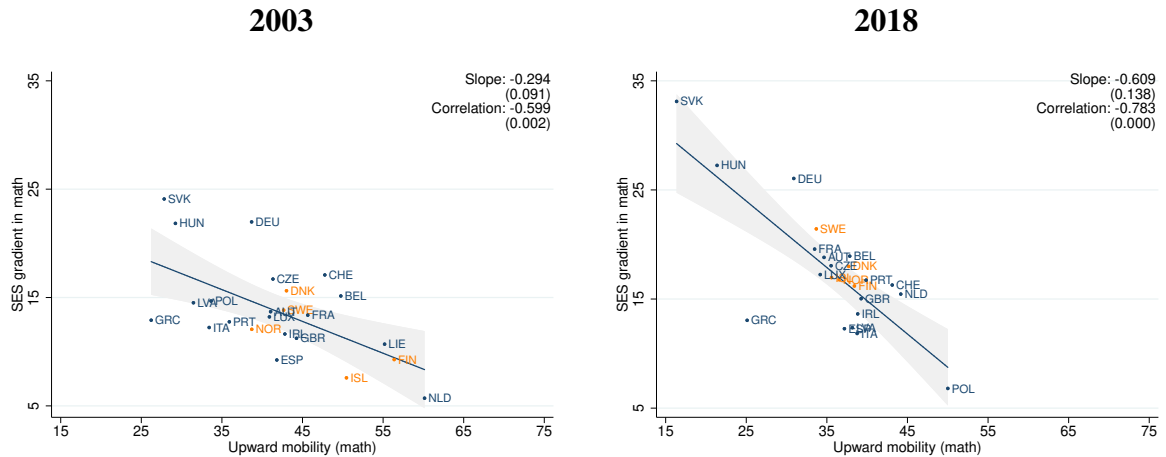
C.1 Upward mobility

Figure C1: Correlation between upward mobility and average math score



Note. The figures present the correlation between upward mobility and average math score. The estimates of upward mobility are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

Figure C2: Correlation between SES gradients in learning and upward mobility (math)



Note. The figures present the correlation between SES gradients in learning and upward mobility (math). The estimates of upward and SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

C.2 Panel estimates: upward mobility in test scores

Table C1: Panel regressions: upward mobility in learning

	Upward mobility (math)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log GDP per capita	8.172 (10.032)						3.302 (9.299)
Returns to secondary school		-0.170* (0.092)					-0.114 (0.211)
Difference between top and lowest tax rates on personal income			0.107 (0.097)				0.114 (0.138)
Government expenditure on education (% of GDP)				-1.898 (1.500)			-1.736 (1.308)
School segregation					-6.633 (21.710)		5.317 (29.794)
SES gradient in parental investment						-0.750*** (0.229)	-0.801** (0.287)
Observations	160	118	137	117	159	142	74
R^2	0.792	0.845	0.815	0.841	0.786	0.874	0.921
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note. The table presents the panel estimates of upward mobility in math scores on economic variables with year and country fixed effects (equation (2)). The estimates of upward mobility and SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Clustered standard errors at the country level (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$).

C.3 SES gradients in other traits

To measure students' educational and occupational aspirations, we follow [La Ferrara \(2019\)](#). Occupational aspiration is defined as a dummy equal to 1 for students who expect to have a job with an International Socio-Economic Index of Occupational Status (ISEI) ≥ 65 at the age of 30 - i.e. a white collar job. Educational aspiration is defined as a dummy equal to 1 for students who expect to study at university. Information on educational aspiration is available from the 2003, 2009, 2015 and 2018 waves, while information on occupational aspiration is available in the 2000, 2003, 2006, 2015 and 2018 waves.

Self-efficacy measures the extent to which individuals believe in their own ability to engage in certain activities and perform specific tasks, especially when facing adverse circumstances ([Bandura, 1977, 1991](#)). PISA measures self-efficacy in math abilities in 2003 and 2012 by combining students' responses to questions on how confident they feel about having to solve certain calculations.

To measure non-cognitive skills, we look at the probability of being on time to school in 2000, 2003, 2012, 2015, and 2018 ([Cunha and Heckman, 2007](#); [Borghans et al., 2008](#)).

It is possible to measure the SES gradient and upward mobility in these traits by estimating equation (1) where the outcome is one of the traits we describe above. Then the SES gradient is the difference in a certain trait between high and low SES and upward mobility is the average outcome of low SES.

Table C2: Correlation among the estimates of SES gradients in other traits in 2003, 2015, 2018

SES gradient in (2003)						
	Math	Read	Science	Educational aspiration	Occupational aspiration	Being on time to school
Math	1					
Read	0.845***	1				
Science	0.859***	0.965***	1			
Educational aspiration	0.458*	0.298	0.314	1		
Occupational aspiration	0.615**	0.461*	0.493*	0.760***	1	
Being on time to school	0.694***	0.692***	0.782***	0.343	0.528**	1

SES gradient in (2015)						
	Math	Read	Science	Educational aspiration	Occupational aspiration	Being on time to school
Math	1					
Read	0.936***	1				
Science	0.973***	0.966***	1			
Educational aspiration	0.125	0.220	0.0938	1		
Occupational aspiration	0.236	0.290	0.192	0.810***	1	
Being on time to school	0.597***	0.640***	0.632***	-0.114	-0.0717	1

SES gradient in (2018)						
	Math	Read	Science	Educational aspiration	Occupational aspiration	Being on time to school
Math	1					
Read	0.962***	1				
Science	0.963***	0.980***	1			
Educational aspiration	0.473**	0.367*	0.345	1		
Occupational aspiration	0.541***	0.386*	0.361*	0.766***	1	
Being on time to school	0.536***	0.565***	0.571***	-0.047	-0.015	1

Note. The table presents the correlation among different SES gradients in other traits in 2003 and 2015. Each measure has been estimated by regressing the outcome of interest on dummy equal to 1 if the mother has at least upper secondary (equation (1)). (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.1$.

Table C3: Correlation among the estimates of upward mobility in other traits in 2003, 2015, 2018

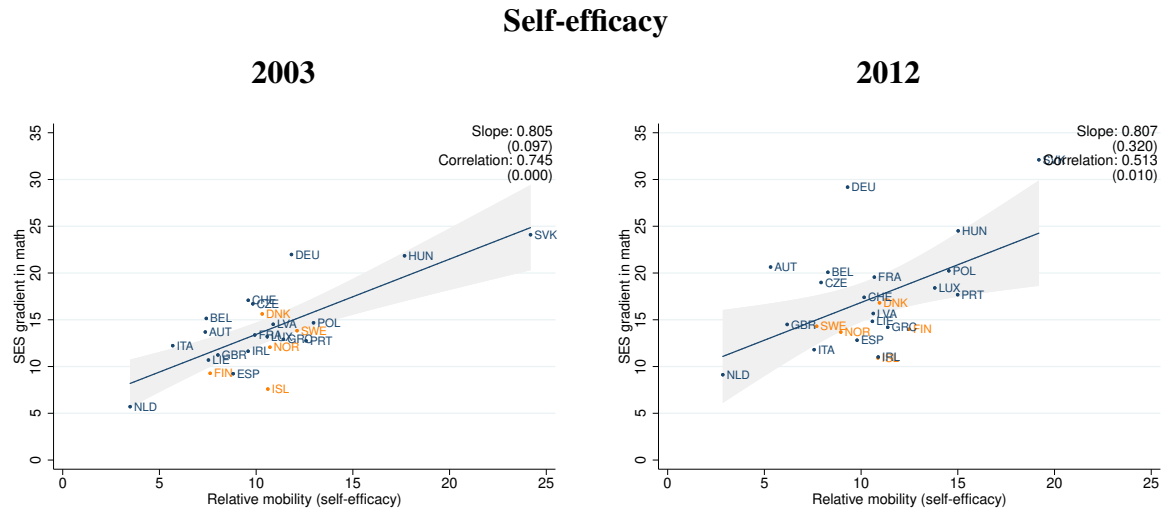
Upward mobility in (2003)						
	Math	Read	Science	Educational aspiration	Occupational aspiration	Being on time to school
Math	1					
Read	0.726***	1				
Science	0.730***	0.832***	1			
Educational aspiration	-0.151	0.370	0.197	1		
Occupational aspiration	0.244	0.732***	0.437	0.720***	1	
Being on time to school	0.262	-0.213	-0.0826	-0.637**	-0.756***	1

Upward mobility in (2015)						
	Math	Read	Science	Educational aspiration	Occupational aspiration	Being on time to school
Math	1					
Read	0.826***	1				
Science	0.887***	0.927***	1			
Educational aspiration	0.131	0.158	0.179	1		
Occupational aspiration	0.239	0.342	0.312	0.316	1	
Being on time to school	0.109	0.288	0.285	-0.0259	0.290	1

Upward mobility in (2018)						
	Math	Read	Science	Educational aspiration	Occupational aspiration	Being on time to school
Math	1					
Read	0.889***	1				
Science	0.934***	0.949***	1			
Educational aspiration	0.416**	0.482**	0.460**	1		
Occupational aspiration	0.610***	0.679***	0.568***	0.570***	1	
Being on time to school	0.196	0.369*	0.253	0.203	0.445**	1

Note. The table presents the correlation among different measures of upward mobility in 2003, 2015 and 2018. Each measure has been estimated by regressing the outcome of interest on dummy equal to 1 if the mother has at least upper secondary (equation (1)). (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.1$.

Figure C3: Correlation between SES gradients in learning (math) and SES gradients in self-efficacy.



Note. The figures present the correlation between SES gradients in learning (math) and SES gradients in self-efficacy. The estimates of the SES gradient in learning are based on equation (1), regression of the rank of the math score on a dummy equal to 1 if the mother has at least upper secondary. The rank is computed at the European level. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the respective p-value in parenthesis.

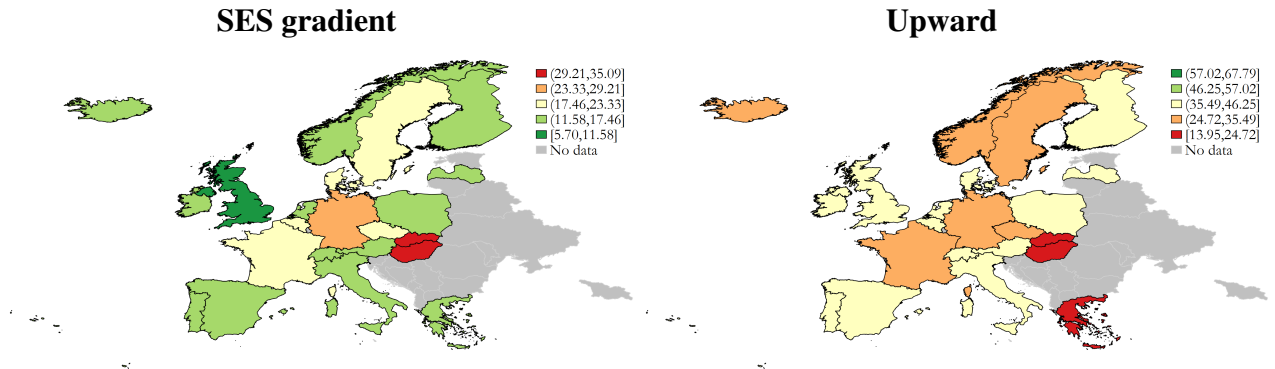
Table C4: SES gradients in other traits and SES gradients in parental investment

	Read	Science	SES gradient in		Being on time	Self-efficacy
			Occupational aspiration	Educational aspiration		
SES gradient in parental investment	0.714*** (0.105)	0.708*** (0.129)	0.009*** (0.001)	0.012*** (0.002)	0.005* (0.002)	0.709*** (0.128)
Observations	24	24	24	24	24	24
R^2	0.492	0.435	0.545	0.486	0.152	0.644

Note. The table presents the correlation between SES gradients in other traits and the socio-economic gradient in parental investment (averaged over the PISA years). The estimates of the SES gradient are based on equation (1), regression of trait on a dummy equal to 1 if the mother has at least upper secondary. Robust standard errors (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$).

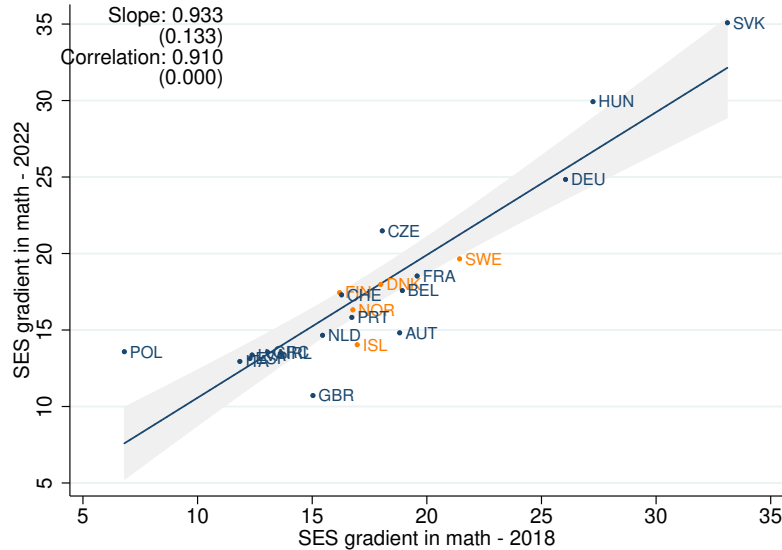
D 2022-PISA wave

Figure D1: Heat map of intergenerational transmission of learning (math score) in 2022



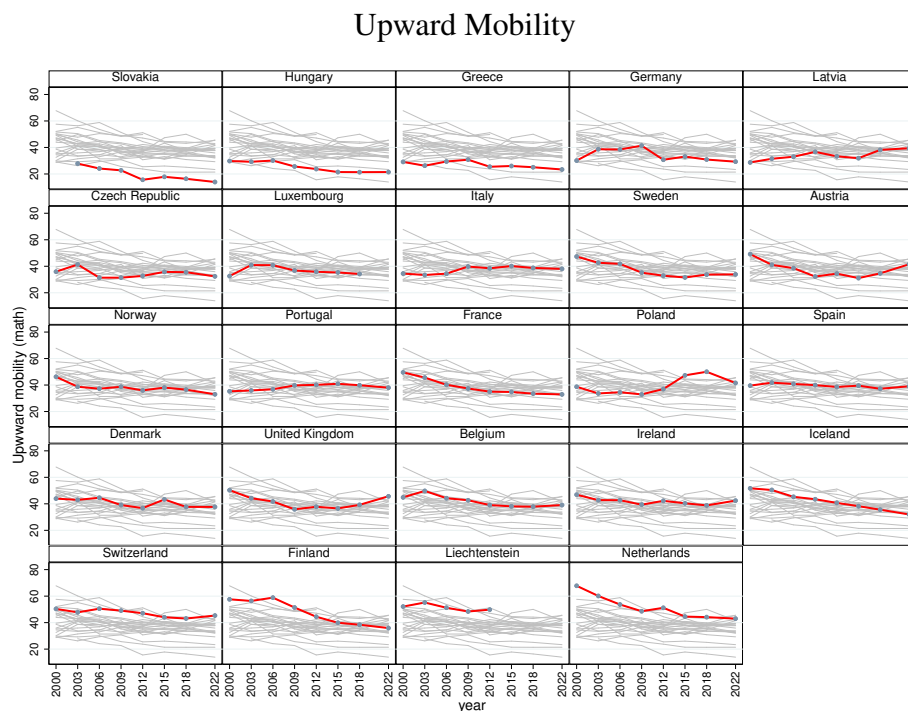
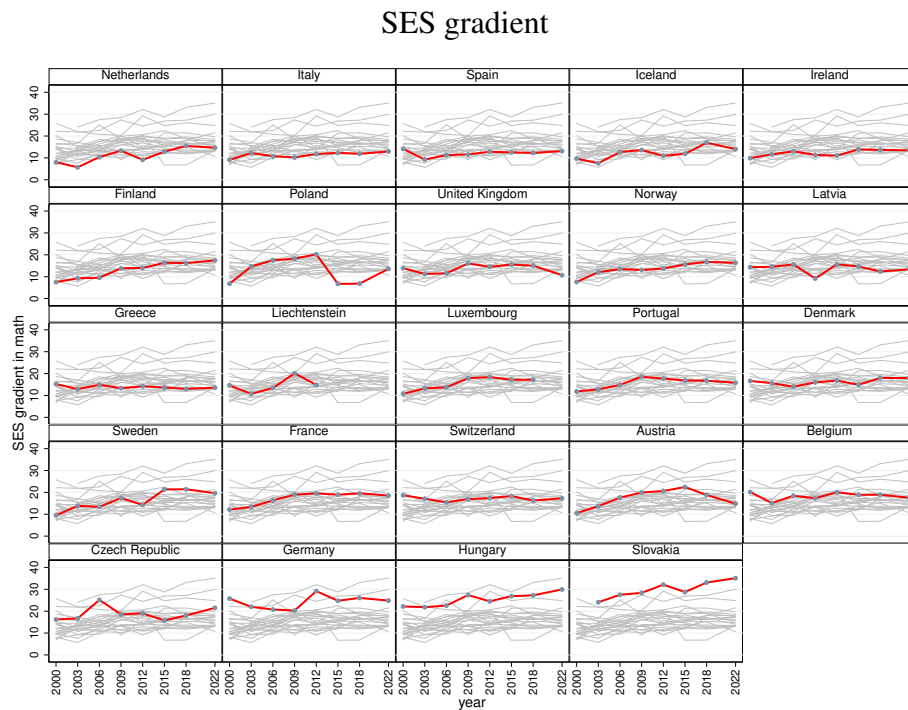
Note. The heat maps present the estimates of intergenerational transmission of learning for each country in 2022 (the estimates are based on equation (1), regression of the rank math score on dummy equal to 1 if the mother has at least upper secondary). The SES gradient in learning measure is β_{tc}^{math} from equation (1). The upward mobility (α_{tc}^{math}) is the average rank of test score of children whose mothers do not have upper secondary. The rank is computed at the European level. Green areas are the most mobile and red areas are the least mobile. The intervals are the same for each year.

Figure D2: Correlation between SES gradient in learning (math) before and after COVID-19 (2018 vs. 2022-PISA wave)



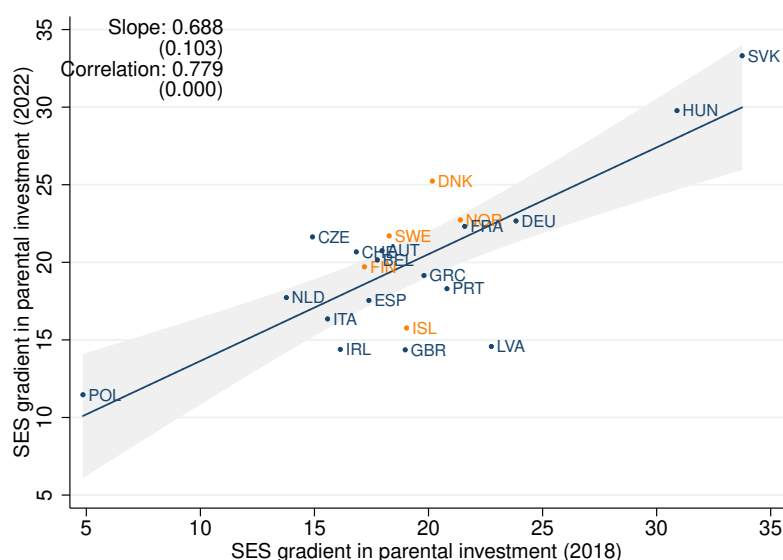
Note. The figure presents the scatter plots between SES gradients in learning (math) before and after COVID-19 (2018 vs. 2022-PISA wave). The estimates of the SES gradient in learning are based on equation (1), regression of the outcome of interest on a dummy equal to 1 if the mother has at least upper secondary. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the p-value in parenthesis.

Figure D3: Trends in intergenerational transmission of learning (math) by country with 2022-PISA wave.



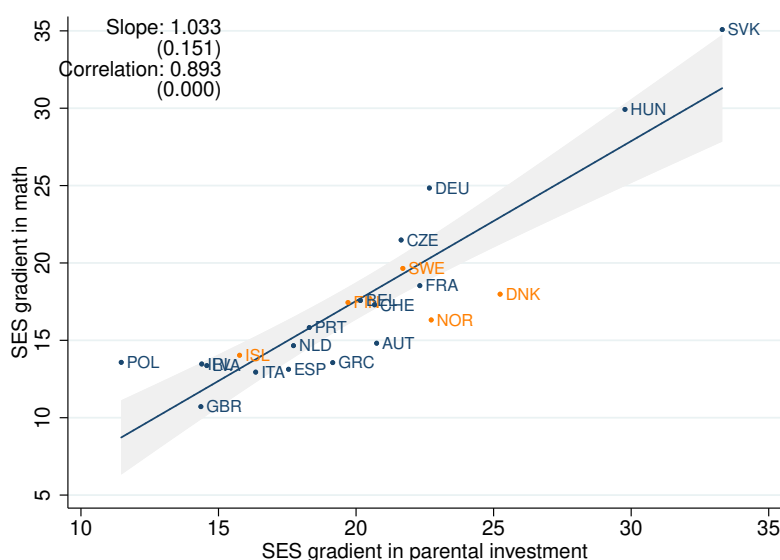
Note. The figures present the trends in the SES gradient and upward mobility for each country with 2022-PISA wave (red line) against other countries (gray lines). The estimates are based on equation (1), regression of the rank math score on dummy equal to 1 if mother has at least completed upper secondary. The rank is computed at the European level. Higher values of the SES gradient in learning correspond to lower mobility. Countries are ordered from the most to least mobile for SES gradient in learning, based on the country-average of SES gradients in learning over the PISA waves. Countries are ordered from the least to the most mobile for upward mobility, based on the country-average of upward mobility over the PISA waves.

Figure D4: Correlation between SES gradients in parental investment before and after COVID-19 (2018 vs. 2022-PISA wave)



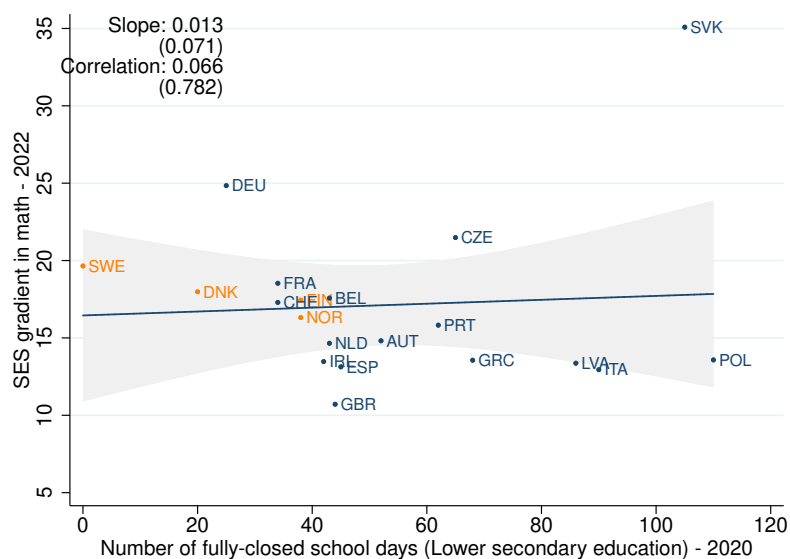
Note. The figure presents the scatter plots between SES gradients in parental investment before and after COVID-19 (2018 vs. 2022-PISA wave). Details on how the SES gradients in parental investment is computed are available in Appendix B.5.1. The estimates of the SES gradient in parental investment are based on equation (1), regression of the outcome of interest on a dummy equal to 1 if the mother has at least upper secondary. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the p-value in parenthesis.

Figure D5: Correlation between SES gradients in learning (math) and SES gradients in parental investment after COVID-19 (2022-PISA wave)



Note. The figure presents the scatter plots between SES gradients in learning (math) and SES gradients in parental investment after COVID-19 (2022 PISA wave). The estimates of the SES gradient in learning are based on equation (1), regression of the outcome of interest on a dummy equal to 1 if the mother has at least upper secondary. Details on how the SES gradients in parental investment is computed are available in Appendix B.5.1. Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the p-value in parenthesis.

Figure D6: Correlation between SES gradients in learning (math) after COVID-19 (2022-PISA wave) and the number of fully-closed school days (Lower secondary education) in 2020



Note. The figure presents the scatter plots between SES gradients in learning (math) after COVID-19 (2022-PISA wave) and the number of fully-closed school days (Lower secondary education) in 2020. The estimates of the SES gradient in learning are based on equation (1), regression of the outcome of interest on a dummy equal to 1 if the mother has at least upper secondary. The number of fully-closed school days (Lower secondary education) in 2020 is from [OECD \(2021\)](#). Confidence interval at 95% level in gray. The notes in the figures show the slope with the respective standard error in parenthesis and the correlation with the p-value in parenthesis.